ENGINEERING CHEMISTRY LAB

Engineering Chemistry Lab is a laboratory course that is designed to provide students with practical knowledge and hands-on experience in the field of chemistry. The course is typically taken by students who are pursuing a degree in engineering or a related field. The main objectives of the course are to:

1. Introduce students to the basic principles and techniques of chemistry.
2. Provide students with the opportunity to conduct experiments and analyze data.
3. Develop students' skills in problem-solving, critical thinking, and communication.
4. Prepare students for advanced courses in chemistry and related fields.

The course typically includes a variety of experiments that cover topics such as:

- Acid-base titrations
- Redox reactions
- Spectrophotometry
- Thermochemistry
- Chemical kinetics
- Electrochemistry
- Organic synthesis

Students are expected to work in teams to conduct experiments, collect and analyze data, and prepare reports. The course may also include lectures, discussions, and demonstrations to supplement the laboratory work.

Overall, the Engineering Chemistry Lab course provides students with a solid foundation in the principles and practices of chemistry, and prepares them for further study and work in the field.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance problem-solving abilities.
4. To encourage independent learning and research.
5. To foster effective communication and collaboration.
6. To prepare students for further academic or professional pursuits.
7. To promote ethical and responsible behavior.
8. To instill a lifelong love of learning.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used for the measurement of physical, chemical, and biological properties of samples. These instruments are used in various fields such as chemistry, biology, physics, and engineering to analyze and identify the composition of materials.

Some fundamental concepts of analytical instruments include:

- **Accuracy**: The degree to which the result of a measurement, calculation, or specification conforms to the correct value or a standard.
- **Precision**: The degree to which repeated measurements under unchanged conditions show the same results.
- **Sensitivity**: The capability of an instrument to respond to small changes in the quantity being measured.
- **Selectivity**: The ability of an instrument to distinguish between different components in a sample.
- **Calibration**: The process of establishing the relationship between the output of an instrument and the value of the input quantity.

Understanding these concepts is essential for the proper use and interpretation of results obtained from analytical instruments.

2. Analysis of Chloride Content, Hardness, and Alkalinity of Water

- **Chloride Content:** Chloride is one of the major anions found in water and wastewater. The chloride content of water can be determined by titration with silver nitrate using potassium chromate as an indicator. This method is known as the Mohr method.

- **Hardness:** Hardness in water is caused by the presence of multivalent metallic cations, such as calcium and magnesium. These cations react with soap to form an insoluble precipitate, which reduces the effectiveness of the soap. Hardness can be determined by titration with a standard solution of ethylenediaminetetraacetic acid (EDTA) using a suitable indicator.

- **Alkalinity:** Alkalinity is a measure of the capacity of water to neutralize acids. It is primarily due to the presence of bicarbonate, carbonate, and hydroxide ions. Alkalinity can be determined by titration with a standard solution of a strong acid, such as sulfuric acid or hydrochloric acid, using a suitable indicator.

These are some of the basic concepts that students should understand about the analysis of chloride content, hardness, and alkalinity of water. It is important to have a good understanding of these concepts in order to accurately analyze and interpret water quality data.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a liquid. It is defined as the negative logarithm of the hydrogen ion concentration in the solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than

7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution.

- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules. Surface tension can be measured using various methods, such as the maximum bubble pressure method or the drop weight method.
- **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. Viscosity can be measured using various methods, such as the capillary tube method or the rotating cylinder method.

These properties are important in various fields, such as chemistry, physics, and engineering. Understanding how to measure and interpret these properties can help students better understand the behavior of liquids in different situations.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. The type of resin being prepared: There are various types of resins, including natural resins, synthetic resins, and modified resins. Each type has its own unique properties and preparation methods.
2. The raw materials used: The raw materials used in the preparation of resins can vary depending on the type of resin being prepared. For example, natural resins are typically derived from plant or animal sources, while synthetic resins are made from chemicals.
3. The processing methods: Different resins require different processing methods, such as heating, cooling, or chemical reactions. These methods can affect the properties of the final resin product.
4. The desired properties of the final resin product: The properties of the final resin product, such as its hardness, flexibility, and chemical resistance, can be influenced by the preparation methods and raw materials used.
5. Safety considerations: The preparation of resins can involve the use of chemicals and high temperatures, so it is important to follow safety guidelines and wear appropriate protective equipment.

By understanding these points, students can gain a better understanding of the preparation of different resins.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol

Adipic acid and paracetamol are two organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the methods of synthesis for both compounds.

Adipic Acid

Adipic acid is a dicarboxylic acid that is commonly used in the production of nylon. It can be synthesized through the following methods:

1. **Conventional Route:** The conventional method of synthesizing adipic acid involves the oxidation of cyclohexane or cyclohexanol with nitric acid. This process produces nitrous oxide, a greenhouse gas, as a byproduct.

2. **Green Route:** The green route for the synthesis of adipic acid involves the use of bio-based feedstocks and more environmentally friendly oxidizing agents. For example, the oxidation of cyclohexene with hydrogen peroxide produces adipic acid without the release of harmful byproducts.

Paracetamol

Paracetamol, also known as acetaminophen, is a common pain reliever and fever reducer. It can be synthesized through the following methods:

1. **Conventional Route:** The conventional method of synthesizing paracetamol involves the nitration of phenol, followed by the reduction of the nitro group to an amine and the acetylation of the amine.

2. **Green Route:** The green route for the synthesis of paracetamol involves the use of more environmentally friendly reagents and processes. For example, the use of microwave irradiation can reduce the reaction time and improve the yield of the reaction.

In conclusion, both adipic acid and paracetamol can be synthesized through conventional and green routes. The green routes offer more environmentally friendly alternatives to the conventional methods. It is important for students to understand the differences between the two routes and the benefits of using green chemistry in the synthesis of organic compounds.

LIST OF EXPERIMENTS

1. **The Milgram Experiment:** A series of social psychology experiments conducted by Yale University psychologist Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts conflicting with their personal conscience.
2. **The Stanford Prison Experiment:** A social psychology experiment conducted at Stanford University in 1971 by Philip Zimbardo, which aimed

to investigate the psychological effects of perceived power, focusing on the struggle between prisoners and prison officers.

3. **The Asch Conformity Experiment:** A series of studies conducted in the 1950s by Solomon Asch, which demonstrated the power of conformity in groups and the influence of majority opinion on individual judgment.
4. **The Pavlov's Dog Experiment:** A series of experiments conducted by Russian physiologist Ivan Pavlov in the 1890s, which demonstrated the principle of classical conditioning, where a neutral stimulus can be used to elicit a conditioned response.
5. **The Marshmallow Test:** A study on delayed gratification conducted by Walter Mischel in the 1960s and 1970s, where children were offered a choice between a small immediate reward or a larger reward if they waited for a short period of time.

1. Calibration of Analytical Equipment and apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

- Calibration should be performed regularly to maintain the accuracy of the instrument.
- The frequency of calibration depends on factors such as the type of instrument, its usage, and the environment in which it is used.
- Calibration can be performed using a reference standard, which is a material or device with a known value.
- The reference standard should be traceable to a national or international standard to ensure its accuracy.
- Calibration can also be performed using a comparison method, where the instrument is compared to another instrument that has been calibrated against a reference standard.
- The results of the calibration should be documented and records should be kept for future reference.
- Any necessary adjustments or repairs should be made to the instrument to ensure its accuracy.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of dissolved calcium and magnesium ions.
- The EDTA (ethylenediaminetetraacetic acid) method is a common method used to determine the hardness of water.
- In this method, a solution of EDTA is used to titrate a water sample.
- The EDTA forms a complex with the calcium and magnesium ions, effectively removing them from the water.
- The end point of the titration is determined using an indicator such as Eriochrome Black T, which changes color when all the calcium and mag-

nesium ions have been complexed.
- The amount of EDTA used in the titration is used to calculate the hardness of the water sample.
- This method is simple, accurate, and widely used in water analysis.

3. Determination of Alkalinity of water sample

Alkalinity is the measure of the water's ability to neutralize acids. It is an important parameter in water quality analysis, as it provides information about the buffering capacity of water. The following are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the sample using a pH meter.
3. If the pH is above 8.3, the sample is considered to have high alkalinity.
4. If the pH is below 4.5, the sample is considered to have low alkalinity.
5. If the pH is between 4.5 and 8.3, titrate the sample with a standard acid solution, such as sulfuric acid or hydrochloric acid, using a burette.
6. Record the volume of acid required to lower the pH to 4.5.
7. Calculate the alkalinity of the sample using the formula: Alkalinity (mg/L as CaCO3) = (Volume of acid used * Normality of acid * 50,000) / Volume of sample.

It is important to note that the alkalinity of water can vary depending on the source and the presence of certain minerals and substances. Therefore, it is recommended to test the alkalinity of water regularly to ensure its quality.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. **Preparation of the solution**: The solution whose pH is to be determined is prepared by dissolving the sample in a suitable solvent.

2. **Selection of an indicator**: An appropriate indicator is chosen based on the pH range of the solution. The indicator changes color at a specific pH, indicating the end point of the titration.

3. **Titration**: A solution of known concentration, called the titrant, is added to the solution being tested. The titrant reacts with the solution, causing a change in pH. The addition of the titrant is continued until the end point is reached, as indicated by the color change of the indicator.

4. **Calculation of pH**: The pH of the solution is calculated using the volume and concentration of the titrant added, and the known reaction between the titrant and the solution.

This method is widely used in analytical chemistry for the determination of the pH of various solutions. It is a simple, accurate, and precise method for pH determination. However, it requires careful selection of the indicator and precise measurement of the volume of the titrant added.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method**: This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: `Surface tension = ( ghr)/2`, where is the density of the liquid, `g` is the acceleration due to gravity, `h` is the height of the liquid column, and `r` is the radius of the capillary tube.

2. **Maximum bubble pressure method**: This method involves measuring the maximum pressure required to blow a bubble from the end of a capillary tube immersed in the liquid. The surface tension is then calculated using the formula: `Surface tension = (4Pm)/d`, where `Pm` is the maximum pressure and `d` is the diameter of the capillary tube.

3. **Drop weight method**: This method involves measuring the weight of a droplet of liquid that detaches from the end of a capillary tube. The surface tension is then calculated using the formula: `Surface tension = mg/ d`, where `m` is the mass of the droplet, `g` is the acceleration due to gravity, and `d` is the diameter of the capillary tube.

4. **Wilhelmy plate method**: This method involves immersing a thin, flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: `Surface tension = F/L`, where `F` is the force and `L` is the perimeter of the plate.

5. **Du Nouy ring method**: This method involves immersing a ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: `Surface tension = (F - Fc)/2 r`, where `F` is the force, `Fc` is the correction factor, and `r` is the radius of the ring.

These are some of the common methods used to determine the surface tension of a given liquid. The choice of method depends on the specific requirements and conditions of the experiment.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the cohesive forces between the molecules. A viscometer is an instrument used to measure the viscosity of a fluid.

Here are the steps to determine the viscosity of a given liquid by viscometer:

1. Fill the viscometer with the liquid to be tested up to the marked level.
2. Place the viscometer in a constant temperature bath to ensure that the temperature of the liquid remains constant during the experiment.
3. Allow the liquid to flow through the capillary tube of the viscometer and measure the time taken for the liquid to flow between two marked points on the viscometer.
4. Repeat the experiment several times to obtain an average value for the time taken for the liquid to flow between the two marked points.
5. Use the Poiseuille equation to calculate the viscosity of the liquid, given the dimensions of the capillary tube, the density of the liquid, and the average time taken for the liquid to flow between the two marked points.

It is important to note that the viscosity of a liquid is dependent on its temperature, so it is essential to maintain a constant temperature during the experiment. Additionally, the viscometer should be calibrated before use to ensure accurate results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate. It is commonly used as a primary standard in redox titrations due to its high purity and stability. The strength of a solution of ferrous ammonium sulfate can be determined using an external indicator, such as potassium permanganate or 1,10-phenanthroline.

1. Potassium permanganate is a strong oxidizing agent that can be used as an external indicator in the titration of ferrous ammonium sulfate. In an acidic solution, potassium permanganate reacts with ferrous ions to form ferric ions and manganese dioxide. The endpoint of the titration is indicated by the appearance of a faint pink color, which indicates that all the ferrous ions have been oxidized.

2. 1,10-phenanthroline is a colorimetric reagent that can be used as an external indicator in the titration of ferrous ammonium sulfate. In the presence of ferrous ions, 1,10-phenanthroline forms a red-colored complex. The endpoint of the titration is indicated by the disappearance of the red color, which indicates that all the ferrous ions have been oxidized.

In summary, the strength of a solution of ferrous ammonium sulfate can be determined using an external indicator such as potassium permanganate or 1,10-phenanthroline. The choice of indicator depends on the specific requirements of the experiment, such as the desired accuracy and precision.

8. Determination of the strength of Potassium dichromate using internal indicator

Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution. The determination of the strength of potassium dichromate using an internal indicator involves the following steps:

1. Preparation of the solution: A known mass of potassium dichromate is dissolved in distilled water to prepare a solution of known strength.

2. Selection of the internal indicator: An internal indicator such as diphenylamine or barium diphenylamine sulfonate is used. The indicator is added to the potassium dichromate solution.

3. Titration: The potassium dichromate solution is titrated against the solution whose strength is to be determined. The end point of the titration is indicated by a color change of the internal indicator.

4. Calculation: The strength of the solution is calculated using the volume of potassium dichromate solution used and the known strength of the potassium dichromate solution.

This method is commonly used in analytical chemistry for the determination of the strength of solutions. It is a simple, accurate, and reliable method.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula Ca(ClO)2. It is widely used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

The available chlorine in bleaching powder can be determined using the following steps:

1. Weigh a sample of bleaching powder and dissolve it in water.
2. Add excess potassium iodide (KI) to the solution. The iodide ions will react with the available chlorine to produce iodine.
3. Titrate the iodine produced with a standard solution of sodium thiosulfate (Na2S2O3) using starch as an indicator. The reaction between iodine and thiosulfate produces iodide ions and sulfate ions.
4. Calculate the amount of available chlorine in the bleaching powder using the volume of thiosulfate solution used in the titration and the concentration of the thiosulfate solution.

This method is known as iodometric titration and is commonly used to determine the available chlorine in bleaching powder. It is important to note that the available chlorine in bleaching powder can decrease over time due to exposure to air and moisture, so it is important to store the powder in a cool, dry place and use it within its shelf life.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method known as the Mohr method.
2. The Mohr method involves titrating the water sample with a silver nitrate solution.
3. The silver nitrate reacts with the chloride ions in the water to form a white precipitate of silver chloride.
4. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate.
5. The amount of silver nitrate used in the titration can be used to calculate the chloride content in the water sample.
6. The Mohr method is a simple and accurate method for determining the chloride content in water.
7. Other methods for determining the chloride content in water include ion chromatography and ion-selective electrodes.
8. It is important to determine the chloride content in water as high levels of chloride can have negative impacts on the environment and human health.
9. Chloride can enter water sources through natural processes such as weathering of rocks and through human activities such as the use of road salt and industrial discharges.
10. Regular monitoring of the chloride content in water is important to ensure that levels remain within safe limits.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed**: Phenol and formaldehyde are mixed in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1.

2. **Formation of hydroxymethylphenols**: The reaction between phenol and formaldehyde produces hydroxymethylphenols, which can further react with phenol to form dimers, trimers, and higher oligomers.

3. **Formation of resin**: The oligomers formed in the previous step can further react with each other to form a three-dimensional network, resulting in the formation of the resin.

4. **Curing**: The resin is then cured by heating it to a high temperature, which causes the resin to harden and become insoluble and infusible.

The properties of the PF resin can be varied by controlling the ratio of formaldehyde to phenol, the type of catalyst used, and the curing conditions. PF resins are widely used in the production of laminates, adhesives, and molded objects. They are known for their high strength, heat resistance, and chemical resistance.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin, also known as urea-methanal, is a non-transparent thermosetting resin or plastic made from urea and formaldehyde. It is commonly synthesized by heating urea and formaldehyde in the presence of a mild base such as ammonia or pyridine.

The most common method of preparation for commercial UF resin is the addition of a second amount of urea during the reaction. The ratio of urea to formaldehyde is between 1:2 and 1:2.2, and therefore methylolation can take place in a short amount of time at temperatures between 90 and 95 °C, with the mixture being maintained under reflux.

One way to prepare UF resin under acidic conditions is to mix 5 g of urea with 6 ml of formalin in a test tube, and shake the test tube until the urea has dissolved. The pH of the solution is then adjusted to 4 by the addition of 4 drops of 0.5N H2SO4, and the time required for precipitation to occur is observed.

Another way to prepare a urea-formaldehyde adhesive is to charge 60 g (1 mole) of urea and 137 g (1.7 mole) of formalin into a 500-mL reaction kettle equipped with a mechanical stirrer and a reflux condenser. The pH of the mixture is adjusted with 2M NaOH to between 7 and 8 as determined by universal indicator paper.

The synthesis of all UF resins can also be done using a typical three-step procedure (alkaline-acidic-alkaline). First, formaldehyde is poured into a three-necked flask and adjusted to pH 7.5 to pH 8.0 with 20% sodium hydroxide solution. The first amount of urea is then added to give the F/U molar ratio of 2.0.

Urea is manufactured from carbon dioxide and ammonia at a temperature of 135–200 °C and at a pressure of 70–230 atmospheres. Formaldehyde is manufactured by the oxidation of methanol, which can be produced from the reaction of carbon dioxide with hydrogen or can be derived from petroleum.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different preparation methods.

Adipic acid:

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is a white crystalline powder and is mainly used in the production of nylon.

1. Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
2. Another method for the preparation of adipic acid is the carbonylation of butadiene to produce adiponitrile, which is then hydrolyzed to adipic acid.

Paracetamol:

Paracetamol, also known as acetaminophen, is an analgesic and antipyretic drug used to relieve pain and reduce fever.

1. Paracetamol can be prepared by the nitration of phenol to produce para-nitrophenol, which is then reduced to para-aminophenol.
2. The para-aminophenol is then acetylated with acetic anhydride to produce paracetamol.

14. Determination of Cell Conductance of a solution

1. Cell conductance is the measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the electrical current passing through the solution.
4. The cell constant is determined by measuring the conductivity of a solution of known conductivity.
5. The cell conductance is then calculated by dividing the measured conductivity by the cell constant.
6. The cell conductance is affected by the concentration, temperature, and type of ions present in the solution.
7. It is important to calibrate the conductivity meter and to measure the temperature of the solution to ensure accurate results.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction of the ester with water.
2. The reaction is typically carried out in the presence of an acid or base catalyst to increase the rate of reaction.
3. The rate of reaction can be measured by monitoring the change in concentration of one of the reactants or products over time.
4. The rate law for the reaction can be determined by measuring the initial rate of reaction at different concentrations of the reactants.
5. The rate constant can then be calculated from the rate law and the measured initial rates of reaction.
6. The rate constant is dependent on temperature and the presence of a catalyst, so these factors must be controlled during the experiment.

7. The rate constant can also be determined by measuring the change in pH of the reaction mixture over time, as the hydrolysis of an ester produces an acid.
8. The rate constant can also be determined by measuring the change in conductivity of the reaction mixture over time, as the hydrolysis of an ester produces ions.
9. The rate constant can also be determined by measuring the change in absorbance of the reaction mixture over time, as the hydrolysis of an ester may produce a colored product.
10. The rate constant can also be determined by measuring the change in volume of the reaction mixture over time, as the hydrolysis of an ester may produce a gas.

16. Element detection and identification of functional groups in organic compounds

Organic compounds are made up of carbon, hydrogen, and other elements such as nitrogen, oxygen, sulfur, and halogens. The detection and identification of these elements and functional groups in organic compounds is an important part of organic chemistry.

1. **Element detection**: Various tests can be used to detect the presence of specific elements in organic compounds. For example, the Lassaigne's test can be used to detect the presence of nitrogen, sulfur, and halogens. The test involves heating the organic compound with sodium metal, which converts the elements into their ionic forms. The resulting sodium fusion extract is then tested for the presence of the ions.

2. **Identification of functional groups**: Functional groups are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of the molecule. Common functional groups include alcohols, amines, carboxylic acids, and ketones. Various chemical tests can be used to identify the presence of specific functional groups in organic compounds. For example, the presence of a carboxylic acid functional group can be confirmed by its reaction with sodium bicarbonate to produce carbon dioxide gas.

In summary, the detection and identification of elements and functional groups in organic compounds is an important part of organic chemistry, and various tests and techniques are available to carry out this task.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided above.
- The instructor also has the option to change any two of the experiments from the list.

- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions for each experiment.
- Students should also be prepared for the possibility that the instructor may change some of the experiments.

Course Outcomes:

1. Understanding of the fundamental concepts and principles of the subject matter.
2. Ability to apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
3. Development of critical thinking and analytical skills.
4. Improvement in communication and collaboration skills.
5. Enhancement of lifelong learning skills and the ability to adapt to new situations and challenges.
6. Acquisition of professional and ethical values.
7. Preparation for further study or career advancement in the field.

Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the knowledge and skills acquired in the course to solve problems.
3. Communicate effectively in written and oral forms.
4. Work effectively in a team.
5. Demonstrate critical thinking and analytical skills.
6. Demonstrate ethical and professional behavior.
7. Demonstrate the ability to learn independently and engage in lifelong learning.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering**: The ability to recall or recognize information.
2. **Understanding**: The ability to comprehend the meaning of information.
3. **Applying**: The ability to use information in a new situation.

4. **Analyzing**: The ability to break down information into parts and understand their relationships.
5. **Evaluating**: The ability to make judgments about the value of information.
6. **Creating**: The ability to combine information to form a new whole.

Course outcomes can be written to target specific levels of Bloom's Taxonomy. For example, an outcome that targets the "applying" level might read: "Students will be able to apply the principles of Newton's Laws to solve problems involving motion." This outcome requires students to use their knowledge of Newton's Laws in a new situation, demonstrating their ability to apply what they have learned.

In summary, course outcomes are statements that describe what students should be able to do by the end of a course, and Bloom's Taxonomy is a framework that can be used to design and assess these outcomes. By aligning course outcomes with the levels of Bloom's Taxonomy, instructors can ensure that their courses are effectively teaching and assessing a range of cognitive skills.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
- When the level is placed on a flat surface, the bubble will settle in the center of the tube, indicating that the surface is level.
- If the bubble is not centered, the surface is not level and needs to be adjusted.
- Levels come in various sizes and shapes, including torpedo, I-beam, and box beam levels.
- They are commonly used in construction, carpentry, and surveying, among other fields.
- Laser levels and digital levels are more advanced types of levels that provide greater accuracy and ease of use.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of samples. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.

3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes**: These instruments magnify small objects to make them visible to the naked eye. They are commonly used to observe the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations. Understanding how to use these instruments and interpret their results is essential for conducting accurate and reliable analyses.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it has an elastic skin, which resists deformation and tends to minimize its surface area.

Some key points to remember when measuring surface tension are:

1. Surface tension is typically measured in units of force per unit length, such as millinewtons per meter (mN/m).
2. There are several methods for measuring surface tension, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
3. The choice of method may depend on factors such as the type of liquid being measured, the accuracy required, and the equipment available.
4. It is important to ensure that the liquid being measured is clean and free of contaminants, as these can affect the surface tension.
5. Temperature can also affect surface tension, so it is important to measure the temperature of the liquid and take this into account when interpreting the results.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.

- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- High quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while sea water is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.

Chloride Content in the Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.
- The conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.

Iron Content in the Water

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- In technology, K3 could refer to a model of a radio transceiver or a keyboard.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-3 Measure the hardness and alkalinity of the water. K3

Water hardness is a measure of the amount of dissolved minerals, primarily calcium and magnesium, in water. Alkalinity, on the other hand, is a measure of the water's ability to neutralize acids and is primarily determined by the presence of bicarbonate, carbonate, and hydroxide ions.

1. **Measuring Water Hardness:** Water hardness can be measured using a variety of methods, including titration, colorimetry, and test strips. The

most common method is titration, where a solution of ethylenediaminetetraacetic acid (EDTA) is used to react with the calcium and magnesium ions in the water sample. The endpoint of the titration is determined using a color indicator, such as Eriochrome Black T, which changes color when all the calcium and magnesium ions have reacted with the EDTA.

2. **Measuring Water Alkalinity:** Water alkalinity can be measured using titration, where a solution of a strong acid, such as sulfuric acid or hydrochloric acid, is added to the water sample until all the bicarbonate, carbonate, and hydroxide ions have reacted. The endpoint of the titration is determined using a pH meter or a color indicator, such as phenolphthalein or bromothymol blue, which changes color at a specific pH.

It is important to measure both the hardness and alkalinity of water as they can affect various water quality parameters, such as pH, corrosion, and scaling. High levels of hardness can cause scaling in pipes and appliances, while high levels of alkalinity can cause corrosion. Additionally, the balance between hardness and alkalinity can affect the effectiveness of water treatment processes, such as disinfection and coagulation.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an aromatic organic compound with the molecular formula C6H5OH. It is also known as carbolic acid, hydroxybenzene, or phenic acid. Here are some fundamental concepts of the preparation of phenol:

1. Phenol can be prepared from the cumene process, which involves the reaction of benzene with propene to form cumene, which is then oxidized to cumene hydroperoxide, followed by acid-catalyzed cleavage to form phenol and acetone.

2. Phenol can also be prepared from the hydrolysis of chlorobenzene, which involves the reaction of chlorobenzene with a strong base, such as sodium hydroxide, to form sodium phenoxide, which is then acidified to form phenol.

3. Another method for the preparation of phenol is the Raschig process, which involves the reaction of benzene with sodium hydroxide and chlorine to form chlorobenzene and sodium chloride, followed by hydrolysis to form phenol.

4. Phenol can also be prepared from the Dow process, which involves the reaction of benzene with propylene oxide to form phenol and acetone.

These are some of the fundamental concepts of the preparation of phenol. It is important to understand these concepts in order to have a good understanding of the preparation of phenol.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- UF is synthesized by the "alkali-acid-alkali" method .
- UF is used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .
- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin for reducing the free formaldehyde.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a type of visa for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- In transportation, K3 may refer to a class of passenger carriages used by the Indian Railways.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k, is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers. The value of k is dependent on the temperature, pressure, and the presence of a catalyst.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Conduct a series of experiments to measure the rate of the reaction at different concentrations of the reactants.
2. Plot the data on a graph with the rate of the reaction on the y-axis and the concentration of the reactants on the x-axis.
3. Determine the order of the reaction with respect to each reactant by analyzing the shape of the graph.
4. Use the integrated rate law for the determined reaction order to calculate the value of the rate constant, k.

It is important to note that the value of k is specific to the reaction conditions and may change if the temperature, pressure, or presence of a catalyst is altered. Therefore, it is necessary to specify the reaction conditions when reporting the value of k.

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- Page 29 is the twenty-ninth page of a forty-page document.
- The content of this page may vary depending on the context of the document.
- It is important to read and understand the content of this page in relation to the rest of the document.
- The information on this page may be crucial for understanding the overall message or purpose of the document.
- It is recommended to take notes and highlight important points while reading this page.
- If there are any questions or confusion regarding the content of this page, it is advised to seek clarification from a knowledgeable source.
- It is important to review this page before moving on to the next page to ensure a thorough understanding of the material.

ENGINEERING PHYSICS LAB KCS

Engineering Physics Lab KCS is a laboratory course designed to provide students with hands-on experience in the application of physics concepts to engineering problems. The course typically covers topics such as:

1. Mechanics: Students perform experiments to study the motion of objects and the forces that cause motion. This includes experiments on projectile motion, Newton's laws of motion, and conservation of energy and momentum.

2. Electricity and Magnetism: Students perform experiments to study the behavior of electric charges and the interaction between electric and magnetic fields. This includes experiments on Coulomb's law, Ohm's law, and electromagnetic induction.

3. Waves and Optics: Students perform experiments to study the behavior of waves, including sound and light waves. This includes experiments on wave interference, diffraction, and polarization.

4. Thermodynamics: Students perform experiments to study the transfer of heat and the relationship between heat and work. This includes experiments on heat engines, heat transfer, and the laws of thermodynamics.

The course is designed to provide students with a strong foundation in the fundamental principles of physics and their application to engineering problems.

Students are expected to work in teams to design and conduct experiments, analyze data, and present their results. The course typically includes both written and oral reports, as well as laboratory notebooks to document experimental procedures and results.

Course Objectives

The course objectives are the specific goals that the course aims to achieve. These objectives are designed to provide students with a clear understanding of what they will learn and what they will be able to do by the end of the course. Some common course objectives include:

1. To provide students with a comprehensive understanding of the subject matter.
2. To develop students' critical thinking and problem-solving skills.
3. To enhance students' ability to communicate effectively, both orally and in writing.
4. To prepare students for further study or professional development in the field.
5. To foster an appreciation for the subject and its relevance to the world around us.

These objectives are typically stated in the course syllabus and are used to guide the development of the course content, assessments, and learning activities. By achieving these objectives, students will have gained the knowledge and skills necessary to succeed in the course and beyond.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, medical diagnosis, and industrial processes. Here are some fundamental concepts of analytical instruments:

1. **Accuracy**: The degree to which the measurement result agrees with the true value of the quantity being measured.
2. **Precision**: The degree to which repeated measurements of the same quantity agree with each other.
3. **Sensitivity**: The smallest change in the quantity being measured that can be detected by the instrument.
4. **Resolution**: The smallest difference between two measurements that can be distinguished by the instrument.
5. **Calibration**: The process of adjusting the instrument to ensure that its measurements are accurate and precise.
6. **Signal-to-Noise Ratio**: The ratio of the strength of the signal being measured to the background noise level.

7. **Dynamic Range**: The range of values over which the instrument can accurately measure the quantity of interest.

These concepts are important to understand when using analytical instruments, as they can affect the accuracy and reliability of the measurements obtained. It is important to calibrate the instrument regularly and to ensure that it is operating within its specified dynamic range to obtain accurate and precise measurements.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content**: Chloride content refers to the amount of chloride ions present in a sample of water. Chloride ions are negatively charged particles that can be found in many types of salts, including sodium chloride (common table salt). The analysis of chloride content in water is important because high levels of chloride can be harmful to aquatic life and can also affect the taste of drinking water.

2. **Hardness**: Water hardness refers to the concentration of calcium and magnesium ions in water. These ions can cause problems in industrial and domestic settings by forming deposits in pipes and appliances. The analysis of water hardness is important to determine the appropriate treatment methods to prevent these problems.

3. **Alkalinity**: Alkalinity is a measure of the ability of water to neutralize acids. It is important to analyze the alkalinity of water because it affects the pH of the water and can influence the effectiveness of water treatment processes. High alkalinity can also cause problems in industrial processes and can affect the taste of drinking water.

In summary, the analysis of chloride content, hardness, and alkalinity is important for understanding the quality of water and determining the appropriate treatment methods. These analyses can help to protect aquatic life, prevent problems in industrial and domestic settings, and ensure that drinking water is safe and palatable.

Water

Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states. It is one of the most plentiful and essential of compounds. A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.

Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen. Water molecules cling to each other because of a force called hydrogen bonding. It's the reason why water can do amazing things. Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid.

Water is the liquid that makes life on Earth possible. As water cycles from the air to the land to the sea and back again, water shapes our planet — and nearly every aspect of our lives. All Living Things Need Water. All living things, from tiny cyanobacteria to giant blue whales, need water to survive.

Water is an inorganic compound with the chemical formula H2O. It is a transparent, tasteless, odorless, and nearly colorless chemical substance, and it is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

Measure of pH, Surface Tension and Viscosity

pH

- pH is a measure of the acidity or basicity of a solution.
- It is defined as the negative logarithm of the hydrogen ion concentration in a solution.
- The pH scale ranges from 0 to 14, with 7 being neutral.
- A pH value less than 7 indicates an acidic solution, while a pH value greater than 7 indicates a basic solution.

Surface Tension

- Surface tension is the property of a liquid that allows it to resist an external force.
- It is caused by the cohesive forces between the molecules of the liquid.
- Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
- The surface tension of a liquid can be affected by various factors, including temperature, impurities, and the presence of surfactants.

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- It is defined as the ratio of the shear stress to the shear rate in a fluid.
- Viscosity is typically measured in units of pascal-seconds ($\mathrm{Pa} \cdot \mathrm{s}$).
- The viscosity of a fluid can be affected by various factors, including temperature, pressure, and the presence of impurities.

Liquid

A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma. It is characterized by the following properties:

1. Liquids have a definite volume, but no definite shape. They take the shape of their container.
2. Liquids can flow and be poured, due to their ability to take the shape of their container.
3. The molecules in a liquid are close together, but not as close as in a solid. They are free to move around, but not as freely as in a gas.
4. Liquids have a higher density than gases, but lower than solids.
5. Liquids can be compressed, but not as much as gases.
6. Liquids have a surface tension, which allows them to form droplets and to "stick" to surfaces.
7. Liquids can dissolve other substances, forming solutions.
8. Liquids can evaporate, turning into a gas, or freeze, turning into a solid.

These properties make liquids useful for a variety of purposes, such as transportation, cleaning, and cooling. Many substances can exist in a liquid state, including water, oil, and alcohol. The study of liquids and their properties is known as fluid mechanics.

Preparation of Different Resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are used in a variety of applications, including adhesives, coatings, and plastics. Here are some key points to understand about the preparation of different resins:

1. **Synthetic resins** are typically prepared through the polymerization of monomers. The specific monomers used, as well as the conditions of the polymerization reaction, can affect the properties of the resulting resin.

2. **Natural resins**, on the other hand, are obtained from plants or animals. For example, amber is a natural resin obtained from the fossilized remains of tree sap, while shellac is a resin obtained from the secretions of the lac insect.

3. **Additives** can be used to modify the properties of resins. For example, plasticizers can be added to increase flexibility, while fillers can be used to improve strength or reduce cost.

4. **Curing** is an important step in the preparation of many resins. This involves the use of heat, radiation, or chemical agents to harden the resin and improve its properties.

5. **Different types of resins** can be prepared for different applications. For example, epoxy resins are commonly used as adhesives, while polyurethane resins are often used in coatings and foams.

In summary, the preparation of different resins involves a variety of techniques and can be tailored to produce resins with specific properties for different applications. It is important to understand the properties of the specific resin being used, as well as the techniques used in its preparation, in order to achieve the desired results.

Synthesis of Adipic Acid

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is a white crystalline powder that is used primarily as a precursor to the production of nylon. Here are some points to understand about the synthesis of adipic acid:

1. Adipic acid is produced from a mixture of cyclohexanol and cyclohexanone, known as "KA oil," through a process called oxidation.
2. The oxidation process involves the use of nitric acid as an oxidizing agent, which converts the KA oil into adipic acid.
3. The reaction takes place in a reactor at a temperature of around 60-70°C and a pressure of around 2-3 atm.
4. The reaction produces nitrogen oxides as a byproduct, which are removed from the reactor through a scrubbing process.
5. The resulting adipic acid is then purified through a series of crystallization and filtration steps to produce a high-purity product.

Acid and Paracetamol by Conventional and Green Route

- An alternative route for the synthesis of N-acetyl-p-aminophenol (paracetamol, PAR) and acetylsalicylic acid (aspirin, ASA) using principles of green chemistry has been proposed. This involves solvent-free synthesis combined with microwave-energy irradiation.
- In 2014, Joncour et al., established a green approach involving direct amidation of hydroquinone to obtain paracetamol with maximum conversion. This eliminates the problems associated with the earlier techniques such as the extra step involved, unwanted ortho-isomerization, and generation of the sulfate salt.
- Acetaminophen can also be synthesized from phenol in three steps. In this synthetic route, the solvent from step two is kept to help maximize atom economy. The first step is an electrophilic aromatic substitution on phenol with nitric acid to create pnitrophenol.

List of Experiments

1. **The Milgram Experiment**: A series of social psychology experiments conducted by Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts that conflicted with their personal conscience.

2. **The Stanford Prison Experiment**: A social psychology experiment conducted by Philip Zimbardo in 1971, which studied the psychological effects of becoming a prisoner or prison guard.
3. **The Asch Conformity Experiment**: A series of studies conducted by Solomon Asch in the 1950s, which demonstrated the power of conformity in groups.
4. **The Pavlov's Dog Experiment**: A classical conditioning experiment conducted by Ivan Pavlov in the 1890s, which demonstrated that a neutral stimulus can elicit a conditioned response when paired with an unconditioned stimulus.
5. **The Marshmallow Test**: A study on delayed gratification conducted by Walter Mischel in the 1960s and 1970s, which found that children who were able to delay gratification and resist the temptation to eat a marshmallow immediately in favor of a larger reward later on, tended to have better life outcomes.

Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with those of a known standard. Calibration is important for ensuring the accuracy and reliability of analytical equipment and apparatus.

1. **Purpose of Calibration:** The purpose of calibration is to ensure that the measurements obtained from analytical equipment and apparatus are accurate and reliable. This is important for obtaining accurate and reliable results in analytical procedures.

2. **Calibration Standards:** Calibration standards are materials or devices with known properties that are used to calibrate analytical equipment and apparatus. These standards can be obtained from national or international organizations, or they can be prepared in-house.

3. **Calibration Procedures:** Calibration procedures vary depending on the type of equipment or apparatus being calibrated. Common calibration procedures include comparison with a known standard, comparison with a reference material, and comparison with a known physical property.

4. **Calibration Frequency:** The frequency of calibration depends on the type of equipment or apparatus, its usage, and the requirements of the analytical procedure. Some equipment and apparatus may require daily calibration, while others may only require calibration once a year.

5. **Calibration Records:** It is important to maintain records of calibration, including the date of calibration, the calibration standard used, the results of the calibration, and any corrective actions taken. These records can

be used to demonstrate the accuracy and reliability of the equipment or apparatus.

6. **Calibration Verification:** Calibration verification is the process of verifying that the calibration of an instrument or apparatus is still valid. This can be done by comparing the measurements obtained from the instrument or apparatus with those of a known standard or by performing a control analysis.

In summary, calibration is an important process for ensuring the accuracy and reliability of analytical equipment and apparatus. It involves the use of calibration standards and procedures, and should be performed at regular intervals. Calibration records should be maintained, and calibration verification should be performed to ensure that the calibration is still valid.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of calcium and magnesium ions.
- EDTA (ethylenediaminetetraacetic acid) is a chelating agent that can bind to these ions and form a complex.
- The EDTA method involves titrating a water sample with a standard solution of EDTA.
- The end point of the titration is determined using a metal ion indicator such as Eriochrome Black T.
- The amount of EDTA used in the titration is proportional to the hardness of the water sample.
- The result is usually expressed in terms of milligrams of calcium carbonate per liter of water (mg CaCO3/L).
- This method is widely used for the determination of water hardness in laboratories and in water treatment plants.

Determination of Alkalinity of Water Sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in water quality analysis, as it provides information about the buffering capacity of water. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter.
3. Add a few drops of phenolphthalein indicator to the water sample. If the pH is above 8.3, the water sample will turn pink.

4. Titrate the water sample with a standard solution of sulfuric acid (H_2SO_4) until the pink color disappears. Record the volume of sulfuric acid used.
5. Calculate the alkalinity of the water sample using the following formula: Alkalinity (mg/L as $CaCO_3$) = (Volume of H_2SO_4 used * Normality of H_2SO_4 * 50,000) / Volume of water sample.

This is a standard method for determining the alkalinity of a water sample. It is important to follow the steps carefully and use the correct reagents and equipment to obtain accurate results.

4. Determination of pH by Titrimetric Method

Titration is a laboratory technique used to determine the concentration of an unknown solution by reacting it with a solution of known concentration. The titrimetric method can be used to determine the pH of a solution.

1. **Principle:** The principle behind the titrimetric method for determining pH is based on the neutralization reaction between an acid and a base. A solution of known concentration, either an acid or a base, is added to the solution being tested until the reaction is complete. The volume of the known solution required to reach the endpoint of the reaction is used to calculate the concentration of the unknown solution, and thus its pH.

2. **Procedure:** To determine the pH of a solution by titration, the following steps are typically followed:

   - A sample of the solution being tested is placed in a flask or beaker.
   - A few drops of an indicator, such as phenolphthalein or bromothymol blue, are added to the solution. The indicator will change color at the endpoint of the reaction.
   - A solution of known concentration, either an acid or a base, is added to the solution being tested using a burette. The solution is added slowly while stirring until the indicator changes color, indicating that the endpoint of the reaction has been reached.
   - The volume of the known solution required to reach the endpoint is recorded.
   - The concentration of the unknown solution, and thus its pH, can be calculated using the volume of the known solution and the balanced chemical equation for the reaction.

3. **Advantages:** The titrimetric method for determining pH is relatively simple and inexpensive. It can be used to determine the pH of a wide range of solutions and is suitable for use in both laboratory and field settings.

4. **Limitations:** The accuracy of the titrimetric method for determining pH is dependent on the accuracy of the known solution and the precision of the volume measurements. The method may not be suitable for solutions

that are highly colored or turbid, as the color change of the indicator may be difficult to observe. Additionally, the method may not be suitable for solutions that contain substances that interfere with the indicator or the reaction between the acid and the base.

Determination of Surface Tension of Given Liquid

Surface tension is the property of a liquid that allows it to resist an external force. It is the result of cohesive forces between the liquid molecules. There are several methods to determine the surface tension of a given liquid, including:

1. **Maximum Bubble Pressure Method:** This method involves measuring the maximum pressure required to blow a bubble from a submerged tube into the liquid. The surface tension is calculated from the maximum pressure and the radius of the tube.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated from the weight of the droplet and the radius of the tube.
3. **Capillary Rise Method:** This method involves measuring the height that a liquid rises in a thin tube due to capillary action. The surface tension is calculated from the height of the liquid rise, the radius of the tube, and the density of the liquid.
4. **Wilhelmy Plate Method:** This method involves immersing a thin, flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is calculated from the force and the perimeter of the plate.
5. **Du Nouy Ring Method:** This method involves immersing a thin, circular ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is calculated from the force and the circumference of the ring.

Each of these methods has its advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and follow the appropriate procedures to obtain accurate results.

Determination of Viscosity of a given liquid by viscometer

1. Viscosity is a measure of a fluid's resistance to flow.
2. A viscometer is an instrument used to measure the viscosity of a liquid.

3. There are several types of viscometers, including capillary, falling ball, and rotational viscometers.
4. The method used to determine the viscosity of a liquid using a viscometer depends on the type of viscometer being used.
5. For example, in a capillary viscometer, the time it takes for a liquid to flow through a narrow tube is measured and used to calculate the viscosity of the liquid.
6. In a falling ball viscometer, the time it takes for a ball to fall through a liquid is measured and used to calculate the viscosity of the liquid.
7. In a rotational viscometer, the torque required to rotate a spindle in a liquid is measured and used to calculate the viscosity of the liquid.
8. The viscosity of a liquid can be affected by factors such as temperature and pressure, so it is important to control these variables during the measurement.
9. The viscosity of a liquid is typically reported in units of poise or centipoise.
10. The viscosity of a liquid can be used to predict its behavior in various applications, such as lubrication and flow through pipes.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is (NH4)2Fe(SO4)2.6H2O.
3. It is commonly used as a primary standard in redox titrations due to its high purity and stability.
4. The strength of a solution of ferrous ammonium sulfate can be determined by titrating it against a standard solution of potassium permanganate (KMnO4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide (K3Fe(CN)6).
6. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where potassium permanganate acts as the oxidizing agent and ferrous ammonium sulfate acts as the reducing agent.
7. The end point of the titration is indicated by the appearance of a blue color, due to the formation of Prussian blue (Fe4[Fe(CN)6]3) when all the ferrous ions have been oxidized to ferric ions.
8. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molarity of the potassium permanganate solution.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a substance.
2. An internal indicator is used to determine the endpoint of the titration.
3. The internal indicator changes color when the reaction is complete, indicating that the titration is finished.
4. The strength of the potassium dichromate can then be calculated using the volume of the titrant and the concentration of the substance being titrated.
5. This method is commonly used in analytical chemistry to determine the strength of a substance.

Determination of Available Chlorine in Bleaching Powder

1. The available chlorine present in a bleaching powder sample is determined iodometrically by treating its solution with an excess of potassium iodide solution in an acidic medium.
2. The liberated iodine (I2) is treated with sodium thiosulphate (Na2S2O3) solution using freshly prepared starch solution as an indicator to be added near the endpoint.
3. The chlorine present in the bleaching powder gets reduced with time.
4. The chlorine that is liberated is called available chlorine, and the calculation of available chlorine is called iodometry.
5. Available chlorine is determined by adding a measured amount of sample solution to an acidified solution of potassium iodide, liberating an equivalent amount of iodine. This free iodine is determined by titration with standardized sodium thiosulfate solution, using a starch indicator endpoint.

Determination of Chloride Content in Water Sample

1. Chloride is one of the major anions found in water and wastewater.
2. The chloride content of water can be determined by titration with silver nitrate solution, using potassium chromate as an indicator.
3. The silver nitrate reacts with the chloride ions to form a white precipitate of silver chloride.
4. The end point of the titration is indicated by the formation of a red-brown color, due to the reaction between excess silver ions and the chromate indicator.

5. The concentration of chloride in the water sample can be calculated from the volume of silver nitrate solution used in the titration.
6. This method is known as the Mohr method and is commonly used for the determination of chloride content in water samples.
7. The accuracy of the method can be affected by the presence of other ions, such as bromide and iodide, which also react with silver nitrate.
8. To overcome this interference, the Volhard method can be used, which involves back-titration with a standard solution of potassium thiocyanate.
9. The chloride content of water is important to monitor, as high levels can indicate pollution from sources such as sewage or industrial waste.
10. The World Health Organization recommends a maximum level of 250 mg/L of chloride in drinking water.

Preparation of Phenol Formaldehyde (PF) Resin

Phenol Formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed**: The first step in the preparation of PF resin is to mix phenol and formaldehyde in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1.

2. **The reaction is heated**: The mixture is then heated to a temperature of around 60-90°C to initiate the reaction. The reaction is exothermic, meaning it releases heat, so the temperature must be carefully controlled to prevent the reaction from getting too hot.

3. **The reaction is allowed to proceed**: The reaction is allowed to proceed for several hours, during which time the formaldehyde molecules react with the phenol molecules to form a complex network of interconnected molecules.

4. **The resin is cooled and solidified**: Once the reaction is complete, the resin is cooled and solidified. The solid resin can then be ground into a powder or used in its liquid form.

5. **The resin is cured**: The final step in the preparation of PF resin is to cure the resin. This is typically done by heating the resin to a temperature of around 150-180°C for several hours. During the curing process, the resin undergoes further chemical reactions that increase its strength and durability.

PF resin is widely used in the production of various products, including laminates, plywood, and molded objects. It is known for its high strength, durability, and resistance to heat and chemicals.

Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin, also known as urea-methanal, is a non-transparent thermosetting resin or plastic. It is made from urea and formaldehyde when they are heated in the presence of a mild base such as ammonia or pyridine.

The most common method of preparation for commercial UF resin is the addition of a second amount of urea during the reaction. The ratio of urea to formaldehyde is between 1:2 and 1:2.2 and therefore methylolation can take place at in a short amount of time at temperatures between 90 and 95 °C, with a mixture being maintained under reflux.

One way to prepare UF resin under acidic conditions is to mix 5 g of urea with 6 ml of formalin in a test tube, and shake the test tube until the urea has dissolved. Adjust the pH of the solution to 4 by addition of 4 drops of 0.5N H2SO4, and observe the time required for precipitation to occur.

Another way to prepare a urea-formaldehyde adhesive is to charge 60 grams (1 mole) of urea and 137 g (1.7 mole) formalin into a 500-mL reaction kettle equipped with a mechanical stirrer and a reflux condenser. The pH of the mixture is adjusted with 2M NaOH to between 7 and 8 as determined by universal indicator paper.

The synthesis of all UF resins can also be a typical three-step procedure (alkaline-acidic-alkaline). First, formaldehyde (F=250 g) is poured into a three-necked flask and adjusted to pH 7.5 to pH 8.0 with 20% sodium hydroxide solution. The first amount of urea (U 1 =94.4 g) is added to give the F/U molar ratio of 2.0.

Urea is manufactured from carbon dioxide and ammonia at a temperature of 135–200 °C and at a pressure of 70–230 atmospheres. Formaldehyde is manufactured by the oxidation of methanol which can be produced from the reaction of carbon dioxide with hydrogen or can be derived from petroleum.

Preparation of Adipic acid / Paracetamol

Adipic acid

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is the most important dicarboxylic acid from an industrial perspective, with about 2.5 billion kilograms of this white crystalline powder being produced annually, mainly as a precursor for the production of nylon.

Preparation

Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway. An alternative method with less environmental impact than the traditional method involves the preparation of adipic acid from cyclohexene oxidation with H2O2 and sodium tungstate. This procedure is considered a green method and involves a series of events that make a low environmental impact.

Paracetamol

Paracetamol is a common pain reliever and fever reducer.

Preparation

Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulfuric acid as a catalyst. It is also available as tablets, which may include calcium stearate or magnesium stearate, cellulose, docusate sodium and sodium benzoate or sodium lauryl sulfate, starch, hydroxypropyl methylcellulose, propylene glycol, sodium starch glycolate, polyethylene glycol and Red #40.

14. Determination of Cell Conductance of a Solution

Cell conductance is a measure of a solution's ability to conduct an electric current. It is determined by the concentration and mobility of ions in the solution. Here are the steps to determine the cell conductance of a solution:

1. Prepare the solution to be tested by dissolving a known amount of solute in a known volume of solvent.
2. Fill the conductivity cell with the solution to be tested.
3. Connect the conductivity cell to a conductivity meter.
4. Turn on the conductivity meter and allow it to stabilize.
5. Record the conductivity reading displayed on the meter.
6. Calculate the cell conductance of the solution using the formula: G = k * C, where G is the cell conductance, k is the cell constant, and C is the conductivity reading.

Note that the cell constant, k, is determined by the geometry of the conductivity cell and must be calibrated before use. The cell constant can be determined by measuring the conductivity of a solution with a known cell conductance and solving for k in the above formula.

Determination of Rate constant of hydrolysis of esters

1. The hydrolysis reaction of an ester in pure water is a slow reaction.

2. When a mineral acid like hydrochloric acid is added, the rate of the reaction is enhanced since the H+ ions from the mineral acid act as the catalyst.
3. The rate constant for the hydrolysis reaction of an ester by dilute acid is $0.6931\times 10^{-3} s^{-1}$.
4. The hydrolysis of esters is catalyzed by either an acid or a base.
5. Acidic hydrolysis is simply the reverse of esterification.
6. The ester is heated with a large excess of water containing a strong-acid catalyst.
7. Like esterification, the reaction is reversible and does not go to completion.
8. The rate constant after calculation from the graphs was approximately 0.003min-1 cm -3 for the 1ml and 2ml of ethyl acetate.
9. When ethyl ethanoate is heated under reflux with a dilute acid such as dilute hydrochloric acid or dilute sulfuric acid, the ester reacts with the water present to produce ethanoic acid and ethanol.

Element detection and identification of functional groups in organic compounds

1. **Element detection** refers to the process of identifying the presence of specific elements in a given compound.
2. **Functional groups** are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of that molecule.
3. There are several methods for detecting elements and identifying functional groups in organic compounds, including:
    - **Combustion analysis**: This method involves burning a sample of the compound and analyzing the resulting products to determine the presence of elements such as carbon, hydrogen, and nitrogen.
    - **Infrared spectroscopy**: This technique uses infrared radiation to identify the presence of specific functional groups within a molecule.
    - **Mass spectrometry**: This method involves ionizing a sample of the compound and analyzing the resulting fragments to determine the presence of specific elements and functional groups.
    - **Nuclear magnetic resonance spectroscopy**: This technique uses the interaction of nuclear spins with a magnetic field to identify the presence of specific functional groups within a molecule.
4. These methods can be used in combination to provide a comprehensive analysis of the elements and functional groups present in a given organic compound.

Instructor's Choice of Experiments

The instructor may choose any 10 experiments from the list provided and may also change any two of the above. Here are some possible experiments that the instructor may choose:

1. Experiment 1
2. Experiment 2
3. Experiment 3
4. Experiment 4
5. Experiment 5
6. Experiment 6
7. Experiment 7
8. Experiment 8
9. Experiment 9
10. Experiment 10

It is important to note that the instructor has the flexibility to change any two of the above experiments to better suit the needs of the class or the curriculum. This allows for a more tailored and effective learning experience for the students.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students are expected to demonstrate upon completion of a course. These outcomes serve as a guide for both instructors and students in terms of what is expected to be achieved by the end of the course.

Some key points to consider when developing course outcomes include:

1. Outcomes should be specific, measurable, and achievable within the scope of the course.
2. Outcomes should be aligned with the overall goals and objectives of the program or curriculum.
3. Outcomes should be written in clear and concise language that is easily understood by students.
4. Outcomes should be regularly reviewed and updated to ensure they remain relevant and accurate.

By clearly defining and communicating course outcomes, instructors can help students understand the purpose and value of the course, and provide a framework for assessing student learning and progress. Additionally, course outcomes can help guide the development of course content, assignments, and assessments, ensuring that all course activities are aligned with the desired learning outcomes.

Upon completion of the course, the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the knowledge and skills acquired in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate effectively in both written and oral forms.

5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior.
7. Engage in continuous learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is commonly used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering**: The ability to recall or retrieve previously learned information.
2. **Understanding**: The ability to comprehend the meaning of material.
3. **Applying**: The ability to use learned material in new and concrete situations.
4. **Analyzing**: The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating**: The ability to make judgments about the value of ideas or materials.
6. **Creating**: The ability to put parts together to form a new whole.

By using Bloom's Taxonomy to design course outcomes, instructors can ensure that their courses are challenging and engaging for students, and that assessments accurately measure student learning. Course outcomes written using Bloom's Taxonomy are specific, measurable, and aligned with the course objectives. They provide a clear roadmap for students to follow as they progress through the course, and help instructors to evaluate the effectiveness of their teaching.

Level

1. A level is a tool used to determine if a surface is horizontal or vertical.
2. It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
3. The tube is mounted in a frame with a flat base or a series of parallel lines.
4. When the level is placed on a surface, the position of the bubble indicates whether the surface is level or not.

5. If the bubble is centered between the lines, the surface is level. If the bubble is off-center, the surface is not level.
6. Levels come in various sizes and shapes, including torpedo, line, and laser levels.
7. They are commonly used in construction, carpentry, and surveying, among other fields.
8. A level can also refer to a degree of proficiency or attainment, as in "She has reached a high level of skill in playing the piano."
9. In video games, a level is a stage or area that must be completed to progress to the next stage.
10. In mathematics, a level set is the set of all points where a given function has a constant value.

CO-1: Understanding the use of different analytical instruments

Analytical instruments are used to measure physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including chemistry, biology, medicine, and environmental science. Some common analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.

3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify unknown compounds and to determine the molecular weight of a compound.

4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes**: These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations, and the choice of instrument depends on the specific needs of the analysis. Understanding the use of these instruments is essential for conducting accurate and reliable analyses.

CO_2: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO_2) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have significantly increased the concentration of CO_2 in the atmosphere, contributing to global warming.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface of a liquid. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO_2, several methods can be used, including:

1. The maximum bubble pressure method: This method involves measuring the pressure required to force a gas bubble through a liquid.
2. The drop weight method: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube.
3. The Wilhelmy plate method: This method involves measuring the force required to detach a thin plate from the surface of a liquid.

These methods can be used to accurately measure the surface tension of CO_2 and other liquids. Understanding the surface tension of CO_2 can provide valuable information about its behavior and interactions with other substances.

Viscosity, Conductance of Solution, Chloride and Iron Content in Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- It is commonly referred to as the thickness of a fluid.
- Viscosity is determined by the internal friction between the molecules of a fluid.
- The unit of viscosity is the pascal-second ($\mathrm{Pa} \cdot \mathrm{s}$).

Conductance of Solution

- Conductance is the ability of a solution to conduct an electric current.
- It is the reciprocal of resistance and is measured in siemens (S).
- The conductance of a solution depends on the concentration and mobility of the ions present in the solution.
- The higher the concentration and mobility of the ions, the higher the conductance of the solution.

Chloride Content in Water

- Chloride is a common anion found in water.
- It is usually present in the form of dissolved salts such as sodium chloride (NaCl) or potassium chloride (KCl).
- High levels of chloride in water can indicate contamination from sources such as road salt, wastewater, or industrial processes.
- The acceptable level of chloride in drinking water is 250 mg/L according to the World Health Organization (WHO).

Iron Content in Water

- Iron is a common element found in water.
- It is usually present in the form of dissolved salts such as ferrous sulfate (FeSO4) or ferric chloride (FeCl3).
- High levels of iron in water can cause staining of clothes and plumbing fixtures, and can affect the taste and odor of water.
- The acceptable level of iron in drinking water is 0.3 mg/L according to the World Health Organization (WHO).

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3, a Belgian-Dutch girl group formed in 1998.
2. K3, a mountain in the Karakoram range, also known as Broad Peak.
3. K3, a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.
4. K3, a type of South Korean high-speed train.
5. K3, a type of tax form used in the United States for reporting income from partnerships, S corporations, and trusts.

CO-3 Measure the hardness and alkalinity of the water

Hardness and alkalinity are important parameters to measure in water. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the water's ability to neutralize acids.

Hardness

1. Hardness can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change

occurs.

2. Another method to measure hardness is by using a hardness test strip, which changes color based on the concentration of calcium and magnesium ions in the water.
3. Hardness can also be measured using an electronic meter, which measures the electrical conductivity of the water and calculates the hardness based on this value.

Alkalinity

1. Alkalinity can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs.
2. Another method to measure alkalinity is by using an alkalinity test strip, which changes color based on the concentration of bicarbonate, carbonate, and hydroxide ions in the water.
3. Alkalinity can also be measured using an electronic meter, which measures the pH of the water and calculates the alkalinity based on this value.

It is important to measure the hardness and alkalinity of water to ensure that it is suitable for its intended use. For example, water with high hardness can cause scaling in pipes and appliances, while water with low alkalinity can be corrosive.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H+) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. **From Haloarenes**: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .

2. **From Benzene Sulphonic Acid**: At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.

3. **From Cumene**: Phenol can be prepared in large quantities by the air oxidation of cumene.

4. **From Diazonium Salts**: Phenol can be prepared in small amounts by the hydrolysis of diazonium salts.

5. **From Coal Tar**: Phenol is extracted from the middle oil fraction for commercial purposes at 443 − 503K.
6. **Raschig's Process**: The reaction of benzene, hydrogen chloride, and air at 523K produces phenol.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

Formaldehyde

Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products. It is used in pressed-wood products, such as particleboard, plywood, and fiberboard; glues and adhesives; permanent-press fabrics; paper product coatings; and certain insulation materials.

Urea Formaldehyde Resin

Urea-formaldehyde (UF), also known as urea-methanal, is a nontransparent thermosetting resin or polymer. It is produced from urea and formaldehyde. These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects. Using formaldehyde and urea as raw materials, a stable urea-formaldehyde resin (UF) is synthesized by the "alkali-acid-alkali" method. Unlike most thermosetting resins, UF often shows the appearance of crystal domains.

Adipic Acid

Melamine, polyvinyl alcohol, and adipic acid dihydrazide as modifiers were added to urea-formaldehyde (UF) resin for reducing the free formaldehyde. The influence of addition amount and addition stage of modifiers on the physicochemical property and free formaldehyde content of UF resin was tested.

Paracetamol

K3

K3 can refer to several different things, including:

1. **Vitamin K3**: Also known as menadione, is a synthetic or artificially produced form of vitamin K. This is unlike the other two forms of vitamin K — vitamin K1, known as phylloquinone, and vitamin K2, called menaquinone. Vitamin K3 may be converted into K2 in your liver.

2. **Schedule K-3**: A form used by the Internal Revenue Service (IRS) in the United States. It is used to report a partner's share of income, deductions, credits, etc. for international tax purposes.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k, is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law of the reaction.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by performing experiments in which the concentration of one reactant is varied while the others are kept constant.
2. Write the rate law for the reaction using the determined orders of reaction.
3. Measure the initial rate of the reaction for several sets of initial concentrations of the reactants.
4. Substitute the initial concentrations and the initial rate into the rate law to obtain a set of simultaneous equations.
5. Solve the simultaneous equations to obtain the value of the rate constant, k.

It is important to note that the rate constant is temperature-dependent and its value will change with changes in temperature. The relationship between the rate constant and temperature is given by the Arrhenius equation.

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- I'm sorry, but I need more information to provide you with the content you are looking for.
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ENGINEERING PHYSICS LAB

Engineering Physics Lab is a laboratory course that is usually taken by engineering students in their first year of study. The course is designed to provide students with hands-on experience in conducting experiments and making observations in the field of physics. The experiments are designed to help students understand the fundamental principles of physics and their applications in engineering.

Some of the experiments that are typically conducted in an Engineering Physics Lab include:

1. Measurement of physical quantities such as length, mass, time, and temperature using various instruments.
2. Study of the properties of matter such as elasticity, surface tension, and viscosity.
3. Study of the behavior of light, including reflection, refraction, and interference.
4. Study of the behavior of electric circuits, including Ohm's law, Kirchhoff's laws, and the use of oscilloscopes.
5. Study of the behavior of magnetic fields, including the use of compasses and the measurement of magnetic fields.

The Engineering Physics Lab provides students with the opportunity to develop their experimental skills and to learn how to analyze and interpret data. The course also helps students to develop their problem-solving skills and to learn how to work effectively in a team. The skills and knowledge gained in the Engineering Physics Lab are essential for success in many engineering disciplines.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance problem-solving abilities.
4. To encourage independent learning and self-motivation.
5. To promote effective communication and collaboration.
6. To foster creativity and innovation.
7. To prepare students for further academic or professional pursuits.
8. To instill ethical and responsible behavior.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in various fields such as chemistry, biology, medicine, and environmental science. Some of the fundamental concepts of analytical instruments include:

- **Accuracy and precision:** Accuracy refers to how close a measurement is to the true value, while precision refers to how close repeated measurements are to each other. Both are important in analytical measurements.

- **Sensitivity and selectivity:** Sensitivity refers to the ability of an instrument to detect small changes in the quantity being measured, while selectivity refers to the ability of an instrument to distinguish between different substances.

- **Calibration:** Calibration is the process of adjusting an instrument to ensure that its measurements are accurate. This is typically done by comparing the instrument's readings to those of a known standard.
- **Signal-to-noise ratio:** The signal-to-noise ratio is a measure of the quality of a measurement. It is the ratio of the signal (the information being measured) to the noise (unwanted variations in the measurement).
- **Types of analytical instruments:** There are many different types of analytical instruments, including spectrophotometers, chromatographs, and mass spectrometers. Each type of instrument has its own unique capabilities and limitations.

Understanding these fundamental concepts is essential for students who wish to use analytical instruments effectively in their studies and research. By mastering these concepts, students will be able to select the appropriate instrument for their needs, calibrate it correctly, and interpret the results accurately.

2. Analysis of Chloride Content, Hardness, and Alkalinity of Water

- **Chloride Content:** Chloride is one of the major anions found in water and wastewater. The chloride content of water can be determined by titration with silver nitrate using potassium chromate as an indicator. This method is known as the Mohr method.
- **Hardness:** Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It is usually expressed in terms of calcium carbonate. Hardness can be determined by titration with a standard solution of ethylenediaminetetraacetic acid (EDTA) using eriochrome black T as an indicator.
- **Alkalinity:** Alkalinity is a measure of the capacity of water to neutralize acids. It is mainly caused by the presence of bicarbonate, carbonate, and hydroxide ions. Alkalinity can be determined by titration with a standard solution of sulfuric acid using phenolphthalein and/or methyl orange as an indicator.

These analyses are important for understanding the quality of water and its suitability for various uses. They can also provide information about the presence of contaminants and the effectiveness of water treatment processes.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 is acidic, while a pH greater than 7 is basic. The pH of a solution can be measured using a pH meter or by using pH indicator paper.

- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method.

These concepts are important for students to understand as they play a crucial role in many scientific and industrial processes. Understanding the measure of pH, surface tension, and viscosity of a liquid can help students better understand the behavior of liquids and their interactions with other substances.

4. To enable the students to understand about the preparation of different resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually derived from plants or synthesized in a laboratory. Here are some points to consider when preparing different resins:

1. **Types of resins**: There are several types of resins, including natural resins, synthetic resins, and modified natural resins. Each type of resin has its own unique properties and preparation methods.
2. **Raw materials**: The raw materials used in the preparation of resins vary depending on the type of resin being prepared. For example, natural resins are derived from plant sources, while synthetic resins are typically derived from petrochemicals.
3. **Preparation methods**: The methods used to prepare resins vary depending on the type of resin being prepared. Some common methods include polymerization, condensation, and esterification.
4. **Safety precautions**: The preparation of resins often involves the use of chemicals and high temperatures. It is important to follow proper safety precautions, such as wearing protective equipment and working in a well-ventilated area.
5. **Quality control**: Quality control is an important aspect of resin preparation. This involves testing the resin to ensure that it meets the desired specifications and standards.

In summary, the preparation of different resins involves understanding the types of resins, the raw materials used, the preparation methods, safety precautions, and quality control. By following these guidelines, students can learn how to prepare different resins effectively.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol by Conventional and Green Route

Adipic acid and paracetamol are two important organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the methods of synthesis for both compounds.

Adipic Acid

Adipic acid is an important dicarboxylic acid used in the production of nylon. It can be synthesized through the following methods:

1. **Conventional Route:** The conventional method of synthesizing adipic acid involves the oxidation of cyclohexane or cyclohexanol with nitric acid. This process produces large amounts of nitrous oxide, a potent greenhouse gas.

2. **Green Route:** The green route for the synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide in the presence of a tungsten catalyst. This method produces water as the only byproduct, making it a more environmentally friendly option.

Paracetamol

Paracetamol is a widely used analgesic and antipyretic drug. It can be synthesized through the following methods:

1. **Conventional Route:** The conventional method of synthesizing paracetamol involves the nitration of phenol to produce para-nitrophenol, which is then reduced to para-aminophenol. The para-aminophenol is then acetylated to produce paracetamol.

2. **Green Route:** The green route for the synthesis of paracetamol involves the direct acetylation of para-aminophenol with acetic anhydride in the presence of a solid acid catalyst. This method produces fewer byproducts and is considered a more environmentally friendly option.

In conclusion, both adipic acid and paracetamol can be synthesized through conventional and green routes. The green routes offer more environmentally friendly options for the synthesis of these important organic compounds.

LIST OF EXPERIMENTS

1. **Milgram Experiment**: A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment**: A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.

3. **Pavlov's Dogs**: A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment**: A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect**: A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment**: A study on the effects of observed aggression on behavior, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment**: A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **The Marshmallow Test**: A study on delayed gratification, conducted by Walter Mischel in the 1960s.
9. **The Invisible Gorilla**: A study on inattentional blindness, conducted by Christopher Chabris and Daniel Simons in 1999.
10. **The Trolley Problem**: A thought experiment in ethics, first introduced by Philippa Foot in 1967.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing it with a reference standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

- Calibration should be performed regularly to maintain the accuracy of the instrument.
- The frequency of calibration depends on factors such as the type of instrument, its usage, and the environment in which it is used.
- Calibration can be performed in-house or by an external calibration service.
- Calibration records should be maintained to demonstrate the accuracy of the instrument over time.
- Calibration involves adjusting the instrument to bring its readings into agreement with the reference standard.
- Calibration can be performed using a variety of methods, depending on the type of instrument and the reference standard used.
- Some common methods of calibration include comparison with a known standard, comparison with a reference material, and comparison with a calibrated instrument.
- Calibration is an essential part of quality control in analytical laboratories and is required by many regulatory bodies.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of dissolved calcium and magnesium ions.
- The EDTA (ethylenediaminetetraacetic acid) method is a common method used to determine the hardness of water.

- In this method, a solution of EDTA is used to titrate a water sample.
- The EDTA binds with the calcium and magnesium ions, forming a complex.
- An indicator such as Eriochrome Black T is used to signal the end point of the titration.
- The amount of EDTA used in the titration is proportional to the hardness of the water sample.
- The result is usually expressed in terms of milligrams of calcium carbonate per liter of water (mg $CaCO_3$/L).
- This method is simple, accurate, and widely used in water analysis.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in determining the quality of water for various uses. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter or pH test strips.
3. If the pH is above 7, the water is alkaline and has the ability to neutralize acids.
4. To determine the exact alkalinity of the water sample, titrate the sample with a standard acid solution, such as sulfuric acid or hydrochloric acid.
5. The volume of acid required to bring the pH of the water sample to a neutral value of 7 is a measure of the alkalinity of the water sample.

These are the basic steps to determine the alkalinity of a water sample. It is important to follow proper procedures and use accurate equipment to obtain reliable results.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the indicator changes color. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Maximum Bubble Pressure Method**: This method involves measuring the maximum pressure required to blow a bubble of air through a liquid. The surface tension is then calculated using the Laplace equation.
2. **Drop Weight Method**: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated using the formula for the weight of a droplet.
3. **Capillary Rise Method**: This method involves measuring the height to which a liquid rises in a thin tube due to capillary action. The surface tension is then calculated using the formula for capillary rise.
4. **Wilhelmy Plate Method**: This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula for the force on a flat plate.
5. **Du Nouy Ring Method**: This method involves immersing a thin, circular ring into a liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula for the force on a circular ring.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and control the experimental conditions to obtain accurate results.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the movement of its molecules. A viscometer is an instrument used to measure the viscosity of a fluid.

Here are the steps to determine the viscosity of a given liquid by viscometer:

1. Fill the viscometer with the liquid to be tested up to the marked level.
2. Place the viscometer in a constant temperature bath to ensure that the temperature of the liquid remains constant during the experiment.

3. Allow the liquid to flow through the capillary tube of the viscometer and measure the time taken for the liquid to flow between two marked points on the tube.
4. Repeat the experiment several times to obtain an average value for the time taken for the liquid to flow between the two marked points.
5. Use the Poiseuille equation to calculate the viscosity of the liquid, given the dimensions of the capillary tube, the density of the liquid, and the average time taken for the liquid to flow between the two marked points.

It is important to note that the viscosity of a liquid is dependent on its temperature, so it is essential to maintain a constant temperature during the experiment. Additionally, the viscometer should be calibrated before use to ensure accurate results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is (NH4)2Fe(SO4)2.6H2O.
3. It is commonly used as a standard solution in redox titrations due to its stability and the ease with which it can be prepared.
4. The strength of a ferrous ammonium sulfate solution can be determined using an external indicator such as potassium permanganate (KMnO4).
5. In this method, a known volume of the ferrous ammonium sulfate solution is titrated against a standard solution of potassium permanganate.
6. The end point of the titration is indicated by the disappearance of the pink color of the permanganate solution.
7. The strength of the ferrous ammonium sulfate solution can then be calculated using the volume and concentration of the potassium permanganate solution and the volume of the ferrous ammonium sulfate solution used in the titration.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a reducing agent.
2. An internal indicator is used to determine the endpoint of the titration.
3. The internal indicator used in this experiment is diphenylamine sulfonate.
4. The diphenylamine sulfonate reacts with the excess dichromate ions to produce a blue color, indicating the endpoint of the titration.
5. The strength of the potassium dichromate solution can be calculated using the volume of the reducing agent used and the stoichiometry of the reaction.
6. The reaction between potassium dichromate and the reducing agent is a redox reaction, where the dichromate ions are reduced to chromium ions

and the reducing agent is oxidized.

7. The balanced chemical equation for the reaction can be used to determine the stoichiometry of the reaction and calculate the strength of the potassium dichromate solution.
8. The strength of the potassium dichromate solution is expressed in terms of molarity or normality, depending on the reaction being studied.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula $Ca(OCl)_2$. It is commonly used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

The available chlorine in bleaching powder can be determined using the following steps:

1. Weigh a sample of bleaching powder and dissolve it in water.
2. Add excess potassium iodide (KI) to the solution. The iodide ions react with the available chlorine to produce iodine.
3. Titrate the iodine produced with a standard solution of sodium thiosulfate ($Na_2S_2O_3$) using starch as an indicator.
4. The amount of sodium thiosulfate used in the titration is proportional to the amount of available chlorine in the bleaching powder sample.

The reaction between iodide ions and available chlorine can be represented by the following equation:

$Cl_2 + 2I^- \rightarrow I_2 + 2Cl^-$

The reaction between iodine and sodium thiosulfate can be represented by the following equation:

$I_2 + 2Na_2S_2O_3 \rightarrow Na_2S_4O_6 + 2NaI$

By using the above equations and the volume of sodium thiosulfate used in the titration, the amount of available chlorine in the bleaching powder sample can be calculated. This method is known as the iodometric method for the determination of available chlorine in bleaching powder.

10. Determination of chloride content in water sample

1. **Introduction:** Chloride is one of the major anions found in water and wastewater. The chloride content of water can be used as an indicator of pollution from human or animal waste, industrial effluents, or saltwater intrusion.
2. **Method:** The most common method for determining the chloride content in water is the argentometric method, also known as the Mohr method.

This method involves titrating the water sample with a standard solution of silver nitrate using potassium chromate as an indicator.

3. **Procedure:** A known volume of the water sample is pipetted into a conical flask and a few drops of potassium chromate indicator are added. The solution is then titrated with a standard solution of silver nitrate until the appearance of a red-brown color, indicating the end point of the titration.
4. **Calculations:** The chloride content of the water sample can be calculated using the volume of silver nitrate solution used and the concentration of the silver nitrate solution.
5. **Interferences:** The presence of other halides, such as bromide and iodide, can interfere with the determination of chloride content using the argentometric method. These interferences can be minimized by using a different indicator or by using a different method for the determination of chloride content.
6. **Accuracy:** The accuracy of the determination of chloride content using the argentometric method depends on the accuracy of the standard solution of silver nitrate and the precision of the titration.
7. **Applications:** The determination of chloride content in water is important in various fields, including environmental monitoring, water treatment, and industrial processes.
8. **Limitations:** The argentometric method for the determination of chloride content in water has some limitations, including interferences from other halides and the need for a skilled operator to perform the titration.
9. **Alternative methods:** Alternative methods for the determination of chloride content in water include ion chromatography, potentiometry, and spectrophotometry.
10. **Conclusion:** The determination of chloride content in water is an important analytical procedure with various applications. The argentometric method is a common and reliable method for the determination of chloride content, but alternative methods are also available.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol or substituted phenol with formaldehyde. PF resins are formed by a step-growth polymerization reaction that can be either acid- or base-catalyzed .

The preparation of PF resin involves mixing phenol, formaldehyde, water, and a catalyst in the desired amount, depending on the resin to be formed, and then heating the mixture. The first part of the reaction, at around 70 °C, forms a thick reddish-brown tacky material, which is rich in hydroxymethyl and benzylic ether groups.

Since formaldehyde exists predominantly in solution as a dynamic equilibrium of methylene glycol oligomers, the concentration of the reactive form of formaldehyde depends on temperature and pH.

PF resins were the first completely synthetic polymers to be commercialized and were used as the basis for Bakelite .

12. Preparation of Urea formaldehyde (UF) resin

Urea-formaldehyde (UF) resin, also known as urea-methanal, is a non-transparent thermosetting resin or plastic made from urea and formaldehyde when they are heated in the presence of a mild base such as ammonia or pyridine.

The most common method of preparation for commercial UF resin is the addition of a second amount of urea during the reaction. The ratio of urea to formaldehyde is between 1:2 and 1:2.2 and therefore methylolation can take place at in a short amount of time at temperatures between 90 and 95 °C, with a mixture being maintained under reflux.

The synthesis of all UF resins is a typical three-step procedure (alkaline-acidic-alkaline). First, formaldehyde is poured into a three-necked flask and adjusted to pH 7.5 to pH 8.0 with 20% sodium hydroxide solution. The first amount of urea is added to give the F/U molar ratio of 2.0.

Urea is manufactured from carbon dioxide and ammonia at a temperature of 135–200 °C and at a pressure of 70–230 atmospheres. Formaldehyde is manufactured by the oxidation of methanol which can be produced from the reaction of carbon dioxide with hydrogen or can be derived from petroleum.

13. Preparation of Adipic acid / Paracetamol

Adipic acid

- Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil.
- The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway.
- Adipic acid is the organic compound with the formula (CH2)4(COOH)2.
- From an industrial perspective, it is the most important dicarboxylic acid: about 2.5 billion kilograms of this white crystalline powder are produced annually, mainly as a precursor for the production of nylon.

Paracetamol

- Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulphuric acid as catalyst.
- It is an effective antipyretic and analgesic.
- Paracetamol is available as 325-mg or 500-mg tablets.

- The tablets may include calcium stearate or magnesium stearate, cellulose, docusate sodium and sodium benzoate or sodium lauryl sulfate, starch, hydroxypropyl methylcellulose, propylene glycol, sodium starch glycolate, polyethylene glycol and Red #40.

14. Determination of Cell Conductance of a solution

1. Cell conductance is a measure of a solution's ability to conduct an electric current.
2. It is determined by measuring the current that flows through a solution between two electrodes.
3. The electrodes are placed at a fixed distance apart and a known potential difference is applied across them.
4. The current that flows through the solution is measured and the cell conductance is calculated using Ohm's law.
5. The cell conductance is dependent on the concentration and mobility of the ions in the solution.
6. It is also affected by the temperature of the solution and the nature of the solvent.
7. The cell conductance can be used to determine the concentration of an electrolyte in a solution.
8. It can also be used to study the properties of ions and their interactions with the solvent.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction between an ester and water.
2. The reaction between an ester and water is known as hydrolysis and results in the formation of a carboxylic acid and an alcohol.
3. The rate of reaction can be measured by monitoring the change in concentration of one of the reactants or products over time.
4. The rate constant can then be calculated using the rate equation for the reaction.
5. The rate equation for the hydrolysis of an ester is given by the expression: rate = kester, where k is the rate constant, [ester] is the concentration of the ester, and water is the concentration of water.
6. By measuring the rate of reaction at different concentrations of the ester and water, the rate constant can be determined by plotting a graph of rate against ester and finding the gradient of the line.
7. The units of the rate constant will depend on the order of the reaction with respect to the ester and water.
8. The order of the reaction can be determined by carrying out initial rate experiments and analyzing the data using the method of initial rates.
9. The rate constant is an important parameter in understanding the kinetics of the hydrolysis of esters and can provide valuable information about the

mechanism of the reaction.

16. Element detection and identification of functional groups in organic compounds

Organic compounds are composed of carbon, hydrogen, and other elements such as nitrogen, oxygen, sulfur, and halogens. The presence of these elements can be detected through various tests.

1. **Detection of Carbon and Hydrogen:** Carbon and hydrogen are detected by heating the organic compound with copper oxide in a test tube. Carbon is oxidized to carbon dioxide and hydrogen is oxidized to water. The carbon dioxide turns lime water milky, while the water is absorbed by anhydrous copper sulfate, turning it blue.

2. **Detection of Nitrogen:** Nitrogen can be detected by the Lassaigne's test. The organic compound is fused with sodium metal, which converts nitrogen into sodium cyanide. The sodium cyanide is then converted into sodium ferrocyanide by reacting with ferrous sulfate. The presence of sodium ferrocyanide is confirmed by the formation of a Prussian blue precipitate when ferric chloride is added.

3. **Detection of Halogens:** Halogens can be detected by the Beilstein test. The organic compound is heated with copper wire, and the flame turns green if halogens are present.

4. **Detection of Sulfur:** Sulfur can be detected by the sodium fusion test. The organic compound is fused with sodium metal, which converts sulfur into sodium sulfide. The sodium sulfide is then reacted with sodium nitroprusside, and the presence of sulfur is confirmed by the formation of a violet color.

5. **Identification of functional groups:** Functional groups are specific arrangements of atoms within an organic molecule that are responsible for its chemical behavior. They can be identified by characteristic chemical reactions. For example, alcohols can be identified by their reaction with Lucas reagent, while carboxylic acids can be identified by their reaction with sodium bicarbonate.

These are some of the methods used for element detection and identification of functional groups in organic compounds. It is important to note that these tests are not always conclusive and further analysis may be required for accurate identification.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided above.

- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions for each experiment.
- Students should also be prepared for the possibility that the instructor may change some of the experiments.

Course Outcomes:

1. Understand the fundamental concepts and principles of the subject.
2. Develop the ability to apply the knowledge and skills acquired in the course to solve problems.
3. Enhance critical thinking and analytical skills.
4. Improve communication skills through written and oral presentations.
5. Develop the ability to work effectively in a team.
6. Foster a sense of curiosity and a desire for lifelong learning.
7. Understand the ethical and social implications of the subject.

Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate ideas and arguments clearly and effectively in both written and oral forms.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior in academic and professional settings.
7. Engage in lifelong learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are often aligned with Bloom's Taxonomy, a framework for categorizing educational goals and objectives.

Bloom's Taxonomy consists of six levels of cognitive learning:

1. **Remembering**: The ability to recall previously learned information.
2. **Understanding**: The ability to comprehend the meaning of the material.
3. **Applying**: The ability to use learned material in new and concrete situations.

4. **Analyzing**: The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating**: The ability to make judgments about the value of ideas or materials.
6. **Creating**: The ability to put parts together to form a new whole.

By aligning course outcomes with Bloom's Taxonomy, instructors can ensure that their course is designed to facilitate student learning and mastery of the material. This can also help instructors to assess student learning and provide feedback to improve student performance.

Level

- A level is a tool used to determine if a surface is horizontal or vertical.
- It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
- When the level is placed on a surface, the bubble will move to the highest point of the tube, indicating if the surface is level or not.
- There are different types of levels, including spirit levels, laser levels, and digital levels.
- Levels are commonly used in construction, carpentry, and surveying, among other fields.
- It is important to use a level when installing or building structures to ensure they are straight and level, which can affect their stability and appearance.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, quality control, and medical diagnosis. Some of the most commonly used analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.

3. **Mass Spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical Instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Thermal Analyzers**: These instruments measure the thermal properties of a sample. They are commonly used to determine the melting point, boiling point, and heat capacity of a substance.

Each of these instruments has its own unique capabilities and limitations. Understanding the use of these instruments is essential for conducting accurate and reliable analyses.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it is covered by an elastic film, which resists deformation and tends to minimize the surface area of the liquid.

Some methods for measuring surface tension include:

1. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to force a gas bubble through a liquid.
2. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube.
3. **Wilhelmy plate method:** This method involves measuring the force required to detach a thin, flat plate from the surface of a liquid.
4. **Du Nouy ring method:** This method involves measuring the force required to detach a thin, circular ring from the surface of a liquid.

These methods can be used to measure the surface tension of various liquids, and the results can provide valuable information about the molecular properties of the liquid and its interactions with other substances.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

1. **Viscosity** is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the cohesive forces between the molecules. The viscosity of formation water is a function of pressure, temperature, and dissolved solids. In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature. Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

2. **Conductance of solution** refers to the ability of a solution to conduct an electric current. The conductivity of water is affected by the presence of dissolved ions, such as chloride ions. The more chloride ions present in the water, the higher the conductivity. In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm. High quality deionized water has a conductivity of about 0.05 S/cm at

25 °C, typical drinking water is in the range of 200–800 S/cm, while sea water is about 50 mS/cm (or 0.05 S/cm).

3. **Chloride content** in water refers to the amount of chloride ions present in the water. Chloride ions are approximately 1.9 percent of the mass of seawater.

4. **Iron content** in water was not found in the search results.

K3

- K3 is a term that can refer to several different things, depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a type of visa for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters of water quality. Hardness refers to the amount of dissolved calcium and magnesium in water, while alkalinity refers to the water's ability to neutralize acids.

1. **Measuring Hardness**: Hardness can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to reach the color change is used to calculate the hardness of the water. Alternatively, hardness test kits are available that use colorimetric methods to determine hardness.

2. **Measuring Alkalinity**: Alkalinity can be measured using a titration method, where a solution of known concentration is added to a water sample until a pH change occurs. The amount of solution required to reach the pH change is used to calculate the alkalinity of the water. Alternatively, alkalinity test kits are available that use colorimetric methods to determine alkalinity.

It is important to measure the hardness and alkalinity of water to ensure that it is suitable for its intended use. For example, water with high hardness can cause scaling in pipes and appliances, while water with low alkalinity can be corrosive.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H+) from the hydroxyl group.

There are several methods for the preparation of phenol, including:

1. From Haloarenes: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. From Benzene Sulphonic Acid: At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
3. From Coal Tar: Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.
4. Raschig's Process: The reaction of benzene, hydrogen chloride and air at 523K produces phenol.
5. From Cumene: Phenol can be prepared in large quantities by the air oxidation of cumene.
6. From Diazonium Salts: Small amounts of phenol can be prepared by the hydrolysis of diazonium salts.
7. From Phenylboronic Acid: Small amounts of phenol can be prepared by the peroxide oxidation of phenylboronic acid.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- **Formaldehyde** is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- **Urea-formaldehyde (UF)**, also known as urea-methanal, is a nontransparent thermosetting resin or polymer. It is produced from urea and formaldehyde.
- UF is synthesized by the "alkali-acid-alkali" method using formaldehyde and urea as raw materials .
- UF often shows the appearance of crystal domains, unlike most thermosetting resins .
- UF resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Melamine, polyvinyl alcohol, and adipic acid dihydrazide can be added to UF resin as modifiers for reducing the free formaldehyde.

K3

1. K3 is a term that can refer to several different things, depending on the context in which it is used.
2. In mathematics, K3 can refer to a K3 surface, which is a complex algebraic surface with certain properties.

3. In music, K3 is a Belgian-Dutch girl group that was formed in 1998.
4. In mountaineering, K3 is the name given to the third highest mountain in the Karakoram range, also known as Broad Peak.
5. In transportation, K3 can refer to a class of passenger carriages used by the Korean State Railway.
6. In biology, K3 can refer to a type of edentulous jaw, which is a jaw without teeth.
7. In military, K3 can refer to the Daewoo Precision Industries K3, which is a light machine gun used by the South Korean military.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by k, is a measure of the speed at which a chemical reaction occurs. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by conducting experiments and analyzing the data.
2. Write the rate law for the reaction using the determined orders of the reactants.
3. Conduct experiments to measure the rate of the reaction at different concentrations of the reactants.
4. Use the data obtained from the experiments and the rate law to calculate the value of the rate constant, k.

It is important to note that the value of the rate constant, k, is dependent on the temperature at which the reaction is carried out. The Arrhenius equation can be used to calculate the value of k at different temperatures.

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- I'm sorry, but I need more information to provide you with a response.
- Can you please specify the subject or the book you are referring to?
- This will help me provide you with the relevant information on page 31 of 40.

ENGINEERING MATHEMATICS LAB KCS

Engineering Mathematics Lab is a course designed to provide students with hands-on experience in applying mathematical concepts and techniques to solve engineering problems. The course typically covers topics such as:

1. Numerical methods: Techniques for approximating solutions to mathematical problems that cannot be solved analytically.

2. Linear algebra: The study of vector spaces and linear transformations, with applications to systems of linear equations.
3. Differential equations: The study of equations involving derivatives, with applications to modeling physical phenomena.
4. Probability and statistics: The study of random events and the analysis of data.

In the lab, students use software tools such as MATLAB or Mathematica to implement and test algorithms for solving problems in these areas. The course may also include projects or assignments that require students to apply mathematical concepts to real-world engineering problems.

Course Objectives

- To provide students with a comprehensive understanding of the subject matter.
- To equip students with the necessary skills and knowledge to apply the concepts learned in real-world situations.
- To encourage critical thinking and problem-solving skills.
- To foster an appreciation for the subject and its relevance in today's world.
- To prepare students for further study or career advancement in the field.
- To promote active engagement and participation in the learning process.
- To assess student progress and provide feedback to facilitate improvement.
- To create a supportive and inclusive learning environment.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in scientific research, medical diagnosis, and industrial processes. Here are some fundamental concepts of analytical instruments:

1. **Accuracy and Precision**: Accuracy refers to how close a measurement is to the true value, while precision refers to how close repeated measurements are to each other. Analytical instruments should be both accurate and precise to provide reliable results.

2. **Sensitivity and Selectivity**: Sensitivity refers to the ability of an instrument to detect small changes in the property being measured, while selectivity refers to the ability of an instrument to distinguish between different substances. Analytical instruments should be both sensitive and selective to provide accurate and reliable results.

3. **Calibration**: Calibration is the process of adjusting an instrument to ensure that its measurements are accurate. This is done by comparing the instrument's readings with a known standard and making any necessary

adjustments.

4. **Signal-to-Noise Ratio**: The signal-to-noise ratio (SNR) is a measure of the quality of a measurement. It is the ratio of the signal strength to the background noise. A high SNR indicates a high-quality measurement, while a low SNR indicates a low-quality measurement.

5. **Types of Analytical Instruments**: There are many different types of analytical instruments, including spectrophotometers, chromatographs, mass spectrometers, and microscopes. Each type of instrument has its own unique capabilities and is used for specific applications.

These are some of the fundamental concepts of analytical instruments that students should understand. By understanding these concepts, students will be better equipped to use and interpret the results of analytical instruments in their studies and future careers.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride content refers to the amount of chloride ions present in a sample of water. It is an important parameter to measure as high levels of chloride can indicate contamination from sources such as industrial waste or sewage. The chloride content of water can be determined using various methods, including titration with silver nitrate or using an ion-selective electrode.

2. **Hardness:** Hardness refers to the concentration of calcium and magnesium ions in water. These ions can cause problems such as scaling in pipes and reduced effectiveness of soap and detergents. Hardness can be measured using various methods, including titration with ethylenediaminetetraacetic acid (EDTA) or using a colorimetric test.

3. **Alkalinity:** Alkalinity refers to the ability of water to neutralize acids. It is an important parameter to measure as it can affect the pH of water and the effectiveness of disinfection processes. Alkalinity can be determined using various methods, including titration with a strong acid or using a colorimetric test.

It is important for students to understand these concepts as they are fundamental to the analysis of water quality. By understanding how to measure and interpret these parameters, students can gain a better understanding of the factors that affect water quality and how to manage them.

Water

Water is a chemical compound with the chemical formula H2O. It is a transparent, tasteless, odorless, and nearly colorless liquid that is the main constituent of Earth's streams, lakes, and oceans, and the fluids of most living organisms.

- Water is vital for all known forms of life, even though it provides no calories or organic nutrients.
- Water is a liquid at standard ambient temperature and pressure, but it often co-exists on Earth with its solid state, ice, and gaseous state, water vapor or steam.
- Water covers 71% of the Earth's surface, mostly in seas and oceans.
- Water has a high heat capacity, meaning it can absorb a lot of heat before its temperature rises.
- Water is a good solvent, dissolving a wide range of substances, which makes it an important medium for chemical reactions.
- Water is essential for agriculture, industry, and domestic use.
- Water is also important for transportation, recreation, and energy production.
- Water is a renewable resource, but its availability can be limited by pollution, overuse, and climate change.

Measure of pH, Surface Tension and Viscosity

pH

- pH is a measure of the acidity or basicity of a solution.
- It is defined as the negative logarithm of the hydrogen ion concentration in a solution.
- The pH scale ranges from 0 to 14, with 7 being neutral.
- A pH value less than 7 indicates an acidic solution, while a pH value greater than 7 indicates a basic solution.

Surface Tension

- Surface tension is the property of a liquid that allows it to resist an external force.
- It is caused by the cohesive forces between the molecules of the liquid.
- Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
- The surface tension of a liquid can be affected by various factors, including temperature, the presence of surfactants, and the nature of the liquid itself.

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.

- It is defined as the ratio of the shear stress to the shear rate in a fluid.
- Viscosity is typically measured in units of pascal-seconds ($\text{Pa} \cdot \text{s}$).
- The viscosity of a fluid can be affected by various factors, including temperature, pressure, and the presence of impurities.

Liquid

A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma. It is characterized by the following properties:

1. Liquids have a definite volume, but no definite shape. They take the shape of their container.
2. Liquids can flow and be poured, due to their ability to take the shape of their container.
3. The molecules in a liquid are close together, but not as close as in a solid. They are free to move around, which allows liquids to flow.
4. Liquids have a higher density than gases, but lower than solids.
5. Liquids can be compressed, but not as much as gases.
6. Liquids can evaporate and turn into a gas, or freeze and turn into a solid, depending on the temperature and pressure.

These properties make liquids useful for a variety of purposes, such as transportation, cleaning, and cooling. Some common liquids include water, oil, and alcohol.

Preparation of Different Resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are used in a variety of applications, including adhesives, coatings, and plastics. Here are some key points to understand about the preparation of different resins:

1. **Synthetic resins** are typically prepared through the polymerization of monomers. The specific monomers used, as well as the conditions of the polymerization reaction, can affect the properties of the resulting resin.

2. **Natural resins**, on the other hand, are obtained from plants or animals. For example, amber is a natural resin obtained from the fossilized remains of tree sap, while shellac is a resin obtained from the secretions of the lac insect.

3. **Additives** can be incorporated into resins to modify their properties. For example, plasticizers can be added to increase flexibility, while fillers can be added to improve strength or reduce cost.

4. **Curing** is an important step in the preparation of many resins. This involves the use of heat, radiation, or chemical agents to crosslink the

polymer chains, resulting in a harder and more durable material.

5. **Different types of resins** can be prepared to meet specific needs. For example, epoxy resins are known for their strong adhesive properties and resistance to chemical and environmental damage, while polyester resins are commonly used in the production of fiberglass-reinforced plastics.

In summary, the preparation of different resins involves a variety of techniques and considerations, depending on the specific type of resin being produced and its intended application. Understanding these factors can help students to better appreciate the versatility and usefulness of resins in various industries.

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid is synthesized from cyclohexene, which is derived from benzene.
2. The first step in the synthesis of adipic acid is the oxidation of cyclohexene to produce cyclohexanone.
3. Cyclohexanone is then further oxidized to produce adipic acid.
4. The oxidation reactions are typically carried out using a strong oxidizing agent such as nitric acid or a mixture of nitric and sulfuric acids.
5. The final product, adipic acid, is a white crystalline solid that is soluble in water and has a melting point of 152°C.

These are the basic steps involved in the synthesis of adipic acid. It is important to note that the specific conditions and reagents used in the synthesis can vary depending on the desired yield and purity of the final product.

Acid and Paracetamol by Conventional and Green Route

Paracetamol, also known as acetaminophen, is a widely used analgesic and antipyretic drug. It is commonly synthesized using conventional methods, but there has been a growing interest in developing green routes for its synthesis.

Conventional Route

The conventional route for the synthesis of paracetamol involves the nitration of phenol to produce p-nitrophenol, which is then reduced to p-aminophenol. The p-aminophenol is then acetylated to produce paracetamol.

Green Route

The green route for the synthesis of paracetamol involves the use of principles of green chemistry, such as solvent-free synthesis combined with microwave-energy irradiation. In 2014, Joncour et al. established a green approach involving the direct amidation of hydroquinone to obtain paracetamol with maximum conversion. This approach eliminates the problems associated with the earlier techniques, such as the extra step involved, unwanted ortho-isomerization, and the generation of the sulfate salt.

List of Experiments

1. **Milgram Experiment**: A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment**: A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs**: A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment**: A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect**: A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment**: A study on the effects of observing aggressive behavior, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment**: A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **Marshmallow Test**: A study on delayed gratification, conducted by Walter Mischel in the 1960s.
9. **Bystander Effect**: A study on the diffusion of responsibility, conducted by John Darley and Bibb Latané in the 1960s.
10. **Rosenhan Experiment**: A study on the validity of psychiatric diagnosis, conducted by David Rosenhan in 1973.

Calibration of Analytical Equipment and Apparatus

Calibration is the process of determining the accuracy of an instrument or apparatus by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

1. Calibration is necessary for all analytical equipment and apparatus, including balances, pipettes, thermometers, and spectrophotometers.

2. Calibration should be performed regularly, according to the manufacturer's recommendations or as required by the laboratory's quality control procedures.
3. Calibration can be performed using certified reference materials, which are materials with known properties that have been certified by a recognized organization.
4. Calibration can also be performed using internal standards, which are materials with known properties that are prepared and used within the laboratory.
5. Calibration should be documented, including the date of calibration, the reference materials or standards used, and the results obtained.
6. Calibration records should be maintained and reviewed regularly to ensure that the instrument is performing within acceptable limits.

Calibration is an essential part of good laboratory practice and is necessary for ensuring the accuracy and reliability of analytical results. It is important to follow the manufacturer's recommendations and the laboratory's quality control procedures when calibrating analytical equipment and apparatus.

2. Determination of Hardness of water sample by EDTA method

The hardness of water can be determined by the amounts of calcium and magnesium ions present in water. One of the methods to determine the hardness of water is by using the EDTA titration method. EDTA stands for Ethylenediaminetetraacetic acid, which can form complexes with metal ions except for alkali metal ions.

The procedure for the determination of hardness of water by EDTA titration is as follows:

1. Take a sample volume of 20ml (V ml).
2. Dilute 20ml of the sample in an Erlenmeyer flask to 40ml by adding 20ml of distilled water.
3. Add 1 mL of ammonia buffer to bring the pH to 10 ± 0.1.
4. Add 1 or 2 drops of the indicator solution.
5. Add EDTA titrant to the sample with vigorous shaking till the wine red color just turns blue.

The hardness of water can be calculated as mg/L and reported as "Total Hardness (EDTA) = ______ mg/L as CaCO3" or as "Calcium Hardness (EDTA) = ______ mg/L as CaCO3".

Determination of Alkalinity of Water Sample

1. Alkalinity is the measure of the ability of water to neutralize acids.

2. It is determined by titrating a water sample with a standard acid solution of known concentration.
3. The endpoint of the titration is determined by a color change of an indicator or by a pH meter.
4. The most commonly used indicator is phenolphthalein, which changes color at a pH of 8.3.
5. The amount of acid required to reach the endpoint is used to calculate the alkalinity of the water sample.
6. Alkalinity is usually expressed in milligrams per liter (mg/L) of calcium carbonate (CaCO3).
7. Alkalinity is important in maintaining the pH balance of water and preventing corrosion in water distribution systems.

4. Determination of pH by Titrimetric Method

Titration is a laboratory technique used to determine the concentration of an unknown solution by reacting it with a solution of known concentration. The titrimetric method can be used to determine the pH of a solution.

1. **Acid-Base Titration:** This method involves the titration of an acidic or basic solution with a solution of known concentration of the opposite nature. The pH of the solution can be determined by monitoring the change in pH during the titration using a pH meter or by using an indicator that changes color at a specific pH.

2. **Neutralization Point:** The point at which the acid and base have completely reacted is called the neutralization point. At this point, the pH of the solution is 7 for strong acids and bases. For weak acids and bases, the pH at the neutralization point is not 7 and can be calculated using the dissociation constant of the acid or base.

3. **Calculating pH:** The pH of the solution can be calculated using the concentration of the acid or base and the dissociation constant. For strong acids and bases, the pH can be calculated using the concentration of the acid or base and the formula pH = -log[H+].

4. **Limitations:** The titrimetric method is not suitable for solutions with very low or very high pH values as the change in pH during the titration may not be significant. It is also not suitable for solutions with very low concentrations of acid or base as the volume of titrant required may be too large.

Determination of Surface Tension of Given Liquid

Surface tension is the energy, or work, required to increase the surface area of a liquid due to intermolecular forces. Since these intermolecular forces vary depending on the nature of the liquid or solutes in the liquid, each solution exhibits differing surface tension properties.

One of the methods to measure surface tension of a liquid is to let the liquid drip. Liquid in a capillary tube forms drops at the bottom of the tube. Another method is the drop number method using a stalagmometer. The formula for the calculation of surface tension of given liquid by this method is `rl = nw×dl / (rw / nl×dw)` .

Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly measured using a device called a viscometer. Here are the steps to determine the viscosity of a given liquid using a viscometer:

1. **Select the appropriate viscometer:** Different types of viscometers are available for different types of liquids and measurement ranges. Choose the one that is suitable for the liquid you are testing.
2. **Calibrate the viscometer:** Before taking any measurements, it is important to calibrate the viscometer to ensure accurate results. Follow the manufacturer's instructions for calibration.
3. **Prepare the sample:** The liquid to be tested should be free of impurities and at a consistent temperature. Pour the liquid into the viscometer, taking care not to introduce any air bubbles.
4. **Take the measurement:** Follow the manufacturer's instructions for taking the measurement. This typically involves timing the flow of the liquid through a narrow tube or capillary.
5. **Calculate the viscosity:** Use the data obtained from the measurement and the calibration to calculate the viscosity of the liquid. The calculation method will vary depending on the type of viscometer used.
6. **Record and report the results:** Record the results of the measurement, including any relevant details such as the temperature of the liquid and the type of viscometer used. Report the results in the appropriate units, typically in centipoise (cP) or millipascal-seconds ($\mathrm{mPa \cdot s}$).

It is important to follow the manufacturer's instructions and standard procedures to ensure accurate and reliable results when determining the viscosity of a liquid using a viscometer.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is (NH4)2Fe(SO4)2.6H2O.
3. It is commonly used as a standard solution in redox titrations due to its stability and the ease of preparation.
4. The strength of a ferrous ammonium sulfate solution can be determined by titrating it against a standard solution of potassium permanganate (KMnO4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide (K3Fe(CN)6).
6. The end point of the titration is indicated by the disappearance of the blue color of the potassium ferricyanide solution.
7. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molarity of the potassium permanganate solution.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is potassium iodide.
4. The reaction between potassium dichromate and potassium iodide produces iodine, which can be detected by its characteristic color.
5. The end point of the titration is reached when all the iodine has been consumed and the solution turns from a yellow color to a colorless solution.
6. The strength of the potassium dichromate solution can be calculated using the volume of the solution used and the known concentration of the iodine solution.
7. This method is accurate and reliable for determining the strength of potassium dichromate solutions.

Determination of Available Chlorine in Bleaching Powder

The determination of available chlorine in bleaching powder is done by the iodometric method. The principle behind this method is that bleaching powder is commonly used as a disinfectant and the chlorine present in the bleaching powder gets reduced with time .

The apparatus required for this experiment includes a mortar and pestle . The reagents required are concentrated glacial acetic acid .

The procedure for this experiment is as follows: 1. Dissolve 1g of bleaching powder in 1 litre of distilled water in a volumetric flask and stopper the container . 2. Available chlorine is determined by adding a measured amount of sample solution to an acidified solution of potassium iodide, liberating an equivalent amount of iodine . 3. This free iodine is determined by titration with standardized sodium thiosulfate solution, using a starch indicator endpoint .

The observation for this experiment is that the bleaching powder solution is titrated with a standard sodium thiosulfate solution (0.025 N) .

Determination of Chloride Content in Water Sample

1. Chloride is one of the major anions found in water and wastewater.
2. The chloride content of water can be determined using a titration method known as the Mohr method.
3. The Mohr method involves titrating the water sample with a solution of silver nitrate (AgNO3) using potassium chromate (K2CrO4) as an indicator.
4. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate (Ag2CrO4).
5. The amount of silver nitrate used in the titration is directly proportional to the chloride content of the water sample.
6. The chloride content can be calculated using the formula: Cl- (mg/L) = (AgNO3 used (mL) x AgNO3 normality (N)) / sample volume (mL).
7. The Mohr method is suitable for determining chloride concentrations in the range of 5 to 1000 mg/L.
8. Other methods for determining chloride content in water include ion chromatography, ion-selective electrodes, and spectrophotometry.
9. It is important to determine the chloride content of water as high levels of chloride can be harmful to aquatic life and can also affect the taste of drinking water.
10. Regular monitoring of chloride levels in water sources is necessary to ensure the safety and quality of water for human consumption and for the protection of aquatic ecosystems.

Preparation of Phenol Formaldehyde (PF) Resin

Phenol Formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed**: Phenol and formaldehyde are mixed in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1.
2. **Formation of hydroxymethylphenols**: The reaction between phenol and formaldehyde produces hydroxymethylphenols, which can further react with phenol to form dimers, trimers, and higher oligomers.
3. **Formation of resin**: The oligomers formed in the previous step can further react with each other to form a three-dimensional network, resulting in the formation of the resin.
4. **Curing**: The resin is then cured by heating it to a high temperature, which causes the resin to harden and become insoluble and infusible.

PF resin is widely used in the production of laminates, plywood, and other wood products due to its high strength, durability, and resistance to heat and chemicals. It is also used in the manufacture of electrical components, such as circuit boards, and in the production of molded products, such as billiard balls and buttons.

Preparation of Urea formaldehyde (UF) resin

Urea-formaldehyde (UF) resin, also known as urea-methanal, is a non-transparent thermosetting resin or plastic made from urea and formaldehyde heated in the presence of a mild base such as ammonia or pyridine .

The most common method of preparation for commercial UF resin is the addition of a second amount of urea during the reaction. The ratio of urea to formaldehyde is between 1:2 and 1:2.2 and therefore methylolation can take place in a short amount of time at temperatures between 90 and 95 °C, with a mixture being maintained under reflux .

One way to prepare UF resin under acidic conditions is to mix 5 g of urea with 6 ml of formalin in a test tube, and shake the test tube until the urea has dissolved. Adjust the pH of the solution to 4 by addition of 4 drops of 0.5N H2SO4, and observe the time required for precipitation to occur .

Another way to prepare a urea-formaldehyde adhesive is to charge sixty grams (1 mole) of urea and 137 g (1.7 mole) formalin into a 500-mL reaction kettle equipped with a mechanical stirrer and a reflux condenser. The pH of the mixture is adjusted with 2M NaOH to between 7 and 8 as determined by universal indicator paper .

The synthesis of all UF resins can be a typical three-step procedure (alkaline-acidic-alkaline). First, formaldehyde (F=250 g) is poured into a three-necked flask and adjusted to pH 7.5 to pH 8.0 with 20% sodium hydroxide solution. The first amount of urea (U 1 =94.4 g) is added to give the F/U molar ratio of 2.0 .

Urea is manufactured from carbon dioxide and ammonia at a temperature of 135–200 °C and at a pressure of 70–230 atmospheres. Formaldehyde is manufactured by the oxidation of methanol which can be produced from the reaction of carbon dioxide with hydrogen or can be derived from petroleum .

Preparation of Adipic acid / Paracetamol

Adipic acid

Adipic acid is an organic compound with the formula (CH2)4(COOH)2. It is the most important dicarboxylic acid from an industrial perspective, with about 2.5 billion kilograms of this white crystalline powder being produced annually, mainly as a precursor for the production of nylon .

Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .

Paracetamol

Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulphuric acid as a catalyst .

Paracetamol is an effective antipyretic and analgesic .

Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current flowing between the electrodes.
4. The cell is filled with the solution to be tested and the conductivity meter measures the current flowing between the electrodes.
5. The cell conductance is calculated by dividing the measured current by the applied voltage and the cell constant.
6. The cell constant is determined by calibrating the conductivity meter using a solution of known conductivity.
7. The cell conductance is reported in units of Siemens per meter (S/m).
8. The cell conductance of a solution is affected by factors such as the concentration and mobility of the ions in the solution, the temperature of the solution, and the presence of other substances in the solution.
9. The cell conductance of a solution can be used to determine the concentration of ions in the solution, to monitor changes in the solution, and to assess the purity of the solution.

10. The determination of cell conductance is an important analytical technique in fields such as chemistry, environmental science, and medicine.

Determination of Rate Constant of Hydrolysis of Esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The reaction between an ester and water to produce a carboxylic acid and an alcohol is known as hydrolysis.
3. The rate of hydrolysis of an ester can be determined by measuring the change in concentration of one of the reactants or products over time.
4. The rate law for the hydrolysis of an ester can be written as: rate = kester, where k is the rate constant.
5. The rate constant, k, can be determined by measuring the initial rate of the reaction at different initial concentrations of the ester and water.
6. The initial rate of the reaction can be determined by measuring the change in concentration of one of the reactants or products over a short period of time at the beginning of the reaction.
7. A graph of the initial rate of the reaction versus the initial concentration of the ester can be plotted, and the slope of the line will give the value of the rate constant, k.
8. The units of the rate constant will depend on the order of the reaction with respect to the ester and water.
9. The order of the reaction can be determined by varying the initial concentration of one of the reactants while keeping the other constant and observing the effect on the initial rate of the reaction.
10. The hydrolysis of esters is typically a first-order reaction with respect to the ester and water, meaning that the rate of the reaction is directly proportional to the concentration of the ester and water.
11. The rate constant of the hydrolysis of an ester can also be determined by measuring the time taken for the reaction to reach a certain extent of completion.
12. The time taken for the reaction to reach a certain extent of completion can be determined by measuring the change in concentration of one of the reactants or products over time and using this information to calculate the time taken for the reaction to reach a certain extent of completion.
13. The rate constant can then be calculated using the integrated rate law for a first-order reaction.
14. The integrated rate law for a first-order reaction is given by: ln[A]t = -kt + ln[A]0, where [A]t is the concentration of the reactant at time t, [A]0 is the initial concentration of the reactant, k is the rate constant, and t is the time.
15. By plotting a graph of ln[A]t versus t, the slope of the line will give the

value of the rate constant, k.

Element detection and identification of functional groups in organic compounds

1. **Element detection**: The detection of elements in organic compounds involves the determination of the presence of certain elements such as carbon, hydrogen, nitrogen, sulfur, and halogens.
2. **Carbon and hydrogen**: The presence of carbon and hydrogen in an organic compound can be detected by heating the compound with copper oxide. The carbon in the compound is oxidized to carbon dioxide, which can be detected by passing the gas through lime water, which turns milky. The hydrogen in the compound is oxidized to water, which can be detected by the use of anhydrous copper sulfate, which turns blue when it comes into contact with water.
3. **Nitrogen**: The presence of nitrogen in an organic compound can be detected by the Lassaigne's test. This involves heating the compound with sodium metal, which converts the nitrogen in the compound to sodium cyanide. The sodium cyanide is then converted to sodium ferrocyanide by the addition of ferrous sulfate. The presence of sodium ferrocyanide can be detected by the formation of a blue precipitate when the solution is acidified with hydrochloric acid and ferric chloride is added.
4. **Sulfur**: The presence of sulfur in an organic compound can be detected by heating the compound with sodium, which converts the sulfur to sodium sulfide. The sodium sulfide can be detected by the formation of a black precipitate when the solution is acidified with hydrochloric acid and lead acetate is added.
5. **Halogens**: The presence of halogens in an organic compound can be detected by the Beilstein test. This involves heating a copper wire in a flame until it is free of any impurities and then dipping the wire into the compound and heating it again in the flame. The presence of halogens is indicated by the formation of a green flame.
6. **Functional groups**: The identification of functional groups in organic compounds involves the use of specific chemical reactions that produce characteristic changes in the compound. For example, the presence of a carboxylic acid functional group can be detected by the formation of a precipitate when the compound is treated with sodium bicarbonate. The presence of an alcohol functional group can be detected by the formation of a chromic acid ester when the compound is treated with chromic acid.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read the list of experiments and be prepared for any changes made by the instructor.
- Students should also communicate with the instructor if they have any questions or concerns about the experiments chosen.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students are expected to demonstrate upon completion of a course. These outcomes serve as a guide for instructors in designing course content and assessments, and for students in understanding the goals of the course.

Some key points to consider when developing course outcomes include:

1. Outcomes should be specific, measurable, and achievable within the scope of the course.
2. Outcomes should align with the overall goals of the program or curriculum.
3. Outcomes should be written in clear, concise language that is easily understood by students.
4. Outcomes should focus on what students will be able to do, rather than what they will know.
5. Outcomes should be reviewed and revised regularly to ensure they remain relevant and up-to-date.

By clearly defining course outcomes, instructors can create a more focused and effective learning experience for their students. Additionally, course outcomes can help students understand the purpose of the course and what they can expect to gain from it. This can lead to increased motivation and engagement, and ultimately, improved learning outcomes.

Upon completion of the course the student should be able to:

- Demonstrate knowledge and understanding of the key concepts, theories, and principles of the subject matter.

- Apply the acquired knowledge and skills to solve problems and make informed decisions.
- Communicate effectively using appropriate language and terminology.
- Work collaboratively with others to achieve common goals.
- Demonstrate critical thinking and analytical skills.
- Conduct research and present findings in a clear and concise manner.
- Demonstrate ethical and professional behavior.
- Engage in continuous learning and professional development.

Course Outcomes Bloom's

Bloom's Taxonomy is a hierarchical classification of the different levels of thinking, and should be applied when creating course objectives. Course objectives are brief statements that describe what students will be expected to learn by the end of the course.

Bloom's taxonomy specifically targets these by seeking to increase knowledge (cognitive domain), develop skills (psychomotor domain), or develop emotional aptitude or balance (affective domain).

Bloom's taxonomy primarily provides instructors with a focus for developing their course learning outcomes. There are a number of reasons why a teacher would want to use Bloom's taxonomy. Initially, it can be used to increase one's understanding of the educational process.

Another key aspect of well-written learning outcomes are verbs at the appropriate course level. Use Bloom's Revised Taxonomy to identify verbs that best align with your course level. Often a course utilizes more than one cognitive level.

Bloom's Taxonomy, originally developed by Benjamin Bloom in 1956, is a model of cognitive skills used to classify educational learning objectives and is a helpful tool for the development of learning outcomes. The process outlined in the taxonomy provides a scaffolding around which instructors can design their course.

Bloom's taxonomy is a powerful tool to help develop learning outcomes because it explains the process of learning: Before you can understand a concept, you must remember it. To apply a concept you must first understand it. In order to evaluate a process, you must have analyzed it.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels are commonly used in construction, carpentry, and surveying.
- The most common type of level is the spirit level, which consists of a sealed, curved glass tube filled with liquid and an air bubble.

- The position of the bubble within the tube indicates whether the surface is level or not.
- Other types of levels include laser levels, water levels, and electronic levels.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding tasks.
- It is important to use a level when installing objects such as shelves, cabinets, and picture frames to ensure that they are straight and level.
- Levels can also be used to determine the slope or grade of a surface, which is important in tasks such as landscaping and drainage.

CO-1: Understanding the use of different analytical instruments

Analytical instruments are used to measure and analyze the physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including chemistry, biology, medicine, and environmental science. Some common analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.

3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.

5. **Microscopes**: These instruments magnify small objects or samples. They are commonly used to observe the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations. Understanding how to use these instruments and interpret their results is essential for conducting accurate and reliable analyses.

CO2: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO2) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have led to an increase in CO2 levels, contributing to global warming.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO2, several methods can be used, including:

1. **Maximum Bubble Pressure Method:** This method involves measuring the maximum pressure required to force a gas bubble through a liquid. The surface tension is calculated from the maximum pressure and the radius of the capillary tube used to form the bubble.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated from the weight of the droplet and the radius of the tube.
3. **Wilhelmy Plate Method:** This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it. The surface tension is calculated from the force and the dimensions of the plate.

These are just a few of the methods that can be used to measure the surface tension of CO2. Each method has its advantages and limitations, and the choice of method will depend on the specific requirements of the experiment.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.

- High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm.

Chloride and Iron Content in the Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.
- The presence of iron in water can affect its taste, color, and odor.
- Iron can also cause staining of laundry and plumbing fixtures.
- Iron content in water can be reduced through various treatment methods, such as ion exchange, oxidation, and filtration.

K3

K3 is a term that can refer to several different things. Here are some of the most common meanings of K3:

1. **Vitamin K3**: Vitamin K3, also known as menadione, is a synthetic or artificially produced form of vitamin K. This is unlike the other two forms of vitamin K — vitamin K1, known as phylloquinone, and vitamin K2, called menaquinone, which occur naturally. Vitamin K3 may be converted into K2 in your liver.

2. **Schedule K-3**: Schedule K-3 is a tax form used by the United States Internal Revenue Service (IRS). It is used to report a partner's share of income, deductions, credits, etc. for international tax purposes.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters that are used to measure the quality of water. Here are some points to consider when measuring the hardness and alkalinity of water:

1. **Hardness** refers to the concentration of calcium and magnesium ions in water. These ions can cause problems in industrial and domestic settings by forming scale deposits in pipes and boilers.

2. **Alkalinity** refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water and is important for maintaining a stable pH.

3. To measure the hardness of water, a titration method can be used. This involves adding a solution of ethylenediaminetetraacetic acid (EDTA) to a water sample until all the calcium and magnesium ions have been complexed. The amount of EDTA used is then used to calculate the hardness of the water.
4. To measure the alkalinity of water, a titration method can also be used. This involves adding a solution of a strong acid, such as hydrochloric acid, to a water sample until the pH reaches a specific value. The amount of acid used is then used to calculate the alkalinity of the water.
5. Both hardness and alkalinity can be expressed in different units, such as milligrams per liter (mg/L) or parts per million (ppm). It is important to use the appropriate units when reporting and comparing results.
6. Regular monitoring of the hardness and alkalinity of water is important for maintaining water quality and preventing problems in industrial and domestic settings. It is also important for aquatic life, as changes in hardness and alkalinity can affect the health of aquatic organisms.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H+) from the hydroxyl group.

There are several methods for the preparation of phenol, including:

1. From Haloarenes: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. From Benzene Sulphonic Acid: Sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide at 573 – 623K. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
3. From Cumene: Phenol can be prepared by the air oxidation of cumene.
4. From Diazonium Salts: Phenol can be prepared by the hydrolysis of diazonium salts.
5. From Coal Tar: Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.
6. Raschig's Process: The reaction of benzene, hydrogen chloride, and air at 523K produces phenol.

Formaldehyde, Urea Formaldehyde Resin, Adipic Acid, and Paracetamol

Formaldehyde

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- It is used in pressed-wood products, such as particleboard, plywood, and fiberboard; glues and adhesives; permanent-press fabrics; paper product coatings; and certain insulation materials.

Urea Formaldehyde Resin

- Urea-formaldehyde (UF), also known as urea-methanal, is a nontransparent thermosetting resin or polymer.
- It is produced from urea and formaldehyde.
- These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Using formaldehyde and urea as raw materials, a stable urea-formaldehyde resin (UF) is synthesized by the "alkali-acid-alkali" method .
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .

Adipic Acid

- Adipic acid dihydrazide can be used as a modifier to reduce free formaldehyde in urea-formaldehyde (UF) resin.
- The influence of the addition amount and addition stage of modifiers on the physicochemical property and free formaldehyde content of UF resin was tested.

Paracetamol

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3 surface: In algebraic geometry, a K3 surface is a complex or algebraic smooth, minimal, complete surface that is regular and has trivial canonical bundle.
2. K3 (band): K3 is a Belgian-Dutch girl group with a Dutch repertoire, whose current line-up is composed of Hanne Verbruggen, Marthe De Pillecyn and Klaasje Meijer, mainly aimed at pre-adolescent children.

3. K3 visa: The K-3 nonimmigrant visa is for the foreign-citizen spouse of a United States (U.S.) citizen.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k, is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers. The value of k is dependent on the temperature, pressure, and the presence of a catalyst.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Conduct a series of experiments to measure the rate of the reaction at different concentrations of the reactants.
2. Plot the data on a graph with the rate of the reaction on the y-axis and the concentration of the reactants on the x-axis.
3. Determine the order of the reaction with respect to each reactant by analyzing the shape of the graph.
4. Use the integrated rate law for the reaction to calculate the value of k.

It is important to note that the value of k is specific to the reaction conditions and may change if the temperature, pressure, or the presence of a catalyst is altered. Therefore, it is necessary to specify the reaction conditions when reporting the value of k.

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- Page 32 is the 32nd page of a 40-page document.
- The content of page 32 is dependent on the context of the document.
- Without further information, it is not possible to provide specific details about the content of page 32.
- It is important to read and understand the content of page 32 in the context of the entire document.
- Page 32 may contain important information, diagrams, or explanations relevant to the topic of the document.
- It is recommended to review the content of page 32 thoroughly to ensure understanding and retention of the material.

ENGINEERING MATHEMATICS LAB

Engineering Mathematics Lab is a course designed to provide students with hands-on experience in applying mathematical concepts and techniques to solve engineering problems. The course typically covers topics such as:

1. Differential Equations: Students learn to solve first and second-order differential equations using analytical and numerical methods.
2. Linear Algebra: Students learn to solve systems of linear equations using matrix methods, and to perform operations such as matrix inversion and eigenvalue decomposition.
3. Probability and Statistics: Students learn to apply statistical methods to analyze data, and to use probability theory to model uncertainty in engineering systems.
4. Numerical Methods: Students learn to use numerical methods to approximate solutions to mathematical problems that cannot be solved analytically.

The course typically involves a combination of lectures, computer-based exercises, and group projects. Students are expected to use mathematical software such as MATLAB or Mathematica to complete assignments and projects.

The goal of the Engineering Mathematics Lab is to provide students with the skills and knowledge needed to apply mathematical methods to solve real-world engineering problems. By the end of the course, students should be able to:

- Formulate mathematical models of engineering systems
- Solve mathematical problems using analytical and numerical methods
- Analyze data using statistical methods
- Communicate mathematical results effectively.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to practical situations.
4. To encourage independent learning and self-motivation.
5. To foster effective communication and collaboration skills.
6. To prepare students for further academic or professional pursuits in the field.
7. To promote ethical and responsible behavior in the discipline.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in various fields such as chemistry, biology, medicine, and environmental science. Some of the fundamental concepts of analytical instruments include:

- **Accuracy and precision:** These terms refer to how close a measurement is to the true value (accuracy) and how consistent the measurements are (precision).

- **Sensitivity and selectivity:** Sensitivity refers to the ability of an instrument to detect small changes in the property being measured, while selectivity refers to the ability of an instrument to distinguish between different substances.
- **Calibration:** This is the process of adjusting an instrument to ensure that its measurements are accurate and precise.
- **Signal-to-noise ratio:** This refers to the ratio of the signal (the measurement of interest) to the noise (unwanted variations in the measurement).
- **Types of analytical instruments:** There are many different types of analytical instruments, including spectrophotometers, chromatographs, mass spectrometers, and microscopes.

These are some of the fundamental concepts of analytical instruments that students should understand. A thorough understanding of these concepts will enable students to effectively use and interpret the results from analytical instruments.

2. Analysis of Chloride Content, Hardness, and Alkalinity of Water

- **Chloride Content:** Chloride content in water is an important parameter to measure as it can indicate the presence of pollution from human or animal waste, industrial effluents, or saltwater intrusion. The chloride content of water can be measured using various methods such as titration with silver nitrate or using ion-selective electrodes.

- **Hardness:** Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It is an important parameter to measure as it can affect the taste of water, the efficiency of soap and detergents, and the scaling of pipes and appliances. Hardness can be measured using various methods such as titration with ethylenediaminetetraacetic acid (EDTA) or using a colorimetric test kit.

- **Alkalinity:** Alkalinity in water is a measure of its ability to neutralize acids. It is an important parameter to measure as it can affect the pH of water and its ability to support aquatic life. Alkalinity can be measured using various methods such as titration with sulfuric acid or using a colorimetric test kit.

These parameters are important to understand and measure in order to ensure the quality and safety of water for various uses. By analyzing the chloride content, hardness, and alkalinity of water, one can determine its suitability for drinking, irrigation, industrial processes, and other applications.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH:** pH is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 is acidic, while a pH greater than 7 is basic. The pH of a solution can be measured using a pH meter or by using pH indicators such as litmus paper.

- **Surface tension:** Surface tension is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity:** Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method. The unit of viscosity is the pascal-second ($Pa \cdot s$).

These concepts are important for understanding the properties of liquids and their behavior in various situations. By understanding these concepts, students can better predict and explain the behavior of liquids in experiments and real-world scenarios.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. The type of resin being prepared: There are various types of resins, including natural resins, synthetic resins, and modified natural resins. Each type of resin has its own specific preparation method.
2. The raw materials used: The raw materials used in the preparation of resins vary depending on the type of resin being prepared. For example, natural resins are obtained from plants, while synthetic resins are prepared from chemicals.
3. The equipment used: The equipment used in the preparation of resins also varies depending on the type of resin being prepared. For example, some resins require heating, while others require mixing or stirring.
4. The processing conditions: The processing conditions, such as temperature, pressure, and time, must be carefully controlled during the preparation of resins to ensure that the desired properties are achieved.
5. Safety precautions: The preparation of resins may involve the use of hazardous chemicals or high temperatures, so it is important to follow appropriate safety precautions to prevent accidents.

In summary, the preparation of different resins involves selecting the appropriate type of resin, raw materials, equipment, and processing conditions, while also following appropriate safety precautions. By understanding these factors, students can learn how to prepare different resins effectively.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol by Conventional and Green Route

Adipic acid and paracetamol are two important organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the synthesis of these compounds through both methods.

Adipic Acid

Adipic acid is an important dicarboxylic acid that is used in the production of nylon. It can be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of adipic acid involves the oxidation of cyclohexane with nitric acid. This process produces nitrous oxide, a greenhouse gas, as a byproduct.

Green Route

The green route for the synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide in the presence of a tungsten catalyst. This process produces water as the only byproduct, making it a more environmentally friendly option.

Paracetamol

Paracetamol is a widely used analgesic and antipyretic drug. It can also be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of paracetamol involves the nitration of phenol to produce 4-nitrophenol, which is then reduced to 4-aminophenol. This is then acetylated to produce paracetamol.

Green Route

The green route for the synthesis of paracetamol involves the direct acetylation of 4-aminophenol with acetic anhydride in the presence of a solid acid catalyst. This process produces fewer byproducts and is considered to be more environmentally friendly.

In conclusion, the synthesis of adipic acid and paracetamol can be achieved through both conventional and green routes. The green routes are considered to be more environmentally friendly due to the production of fewer byproducts and the use of less harmful reagents. It is important for students to understand the differences between these routes and the benefits of using green chemistry in the synthesis of organic compounds.

LIST OF EXPERIMENTS

1. **The Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **The Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **The Asch Conformity Experiment:** A study on the effects of group pressure on individual behavior, conducted by Solomon Asch in the 1950s.
4. **The Pavlov's Dog Experiment:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
5. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.
6. **The Bobo Doll Experiment:** A study on the effects of observed aggression on behavior, conducted by Albert Bandura in the 1960s.
7. **The Hawthorne Effect:** A study on the effects of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
8. **The Bystander Effect:** A study on the diffusion of responsibility in emergency situations, conducted by John Darley and Bibb Latané in the 1960s.
9. **The Blue Eyes/Brown Eyes Experiment:** A study on the effects of discrimination, conducted by Jane Elliott in 1968.
10. **The Little Albert Experiment:** A study on the acquisition of phobias, conducted by John B. Watson and Rosalie Rayner in 1920.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with those of a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure the accuracy of the instrument over time.
- The frequency of calibration will depend on factors such as the type of instrument, its usage, and the environment in which it is used.

- Calibration should be performed by trained personnel using appropriate standards and procedures.
- Records of calibration should be maintained to provide evidence of the instrument's accuracy and to track any changes over time.
- If an instrument is found to be out of calibration, it should be adjusted or repaired before being used again.

In summary, calibration is an essential step in ensuring the accuracy and reliability of analytical equipment and apparatus. It should be performed regularly by trained personnel using appropriate standards and procedures. Records of calibration should be maintained to provide evidence of the instrument's accuracy and to track any changes over time. If an instrument is found to be out of calibration, it should be adjusted or repaired before being used again.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of dissolved calcium and magnesium ions.
- Ethylenediaminetetraacetic acid (EDTA) is a chelating agent that can be used to determine the hardness of water.
- In this method, a solution of EDTA is added to a water sample until all the calcium and magnesium ions have formed complexes with the EDTA.
- The amount of EDTA used is then measured and used to calculate the hardness of the water sample.
- The end point of the titration is determined using an indicator such as Eriochrome Black T, which changes color when all the calcium and magnesium ions have been complexed.
- This method is simple, accurate, and widely used for the determination of water hardness.

3. Determination of Alkalinity of water sample

Alkalinity is the measure of the water's ability to neutralize acids. It is an important parameter in determining the water's buffering capacity, which is the water's ability to resist changes in pH. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter or pH test strips.
3. If the pH is above 7, the water is alkaline. If the pH is below 7, the water is acidic.
4. To determine the exact alkalinity of the water sample, titrate the sample with a standard acid solution, such as sulfuric acid or hydrochloric acid, using a burette.
5. The endpoint of the titration is reached when the pH of the water sample drops to a predetermined value, usually around 4.5.

6. The volume of the acid solution used to reach the endpoint is used to calculate the alkalinity of the water sample.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the color change occurs. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: `T = (h gr)/2cos` , where `T` is the surface tension, `h` is the height of the liquid rise, is the density of the liquid, `g` is the acceleration due to gravity, `r` is the radius of the capillary tube, and is the contact angle between the liquid and the tube.

2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from the end of a capillary tube immersed in the liquid. The surface tension is then calculated using the formula: `T = (4PmR)/(3R + r)`, where `T` is the surface tension, `Pm` is the maximum pressure, `R` is the radius of the bubble, and `r` is the radius of the capillary tube.

3. **Drop weight method:** This method involves measuring the weight of drops of liquid that detach from the end of a capillary tube. The surface

tension is then calculated using the formula: $T = mg/ d$, where T is the surface tension, m is the mass of the drop, g is the acceleration due to gravity, and d is the diameter of the capillary tube.

4. **Wilhelmy plate method:** This method involves immersing a thin, flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: $T = F/Lcos$, where T is the surface tension, F is the force required to detach the plate, L is the perimeter of the plate, and   is the contact angle between the liquid and the plate.

5. **Du Nouy ring method:** This method involves immersing a ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: $T = (F - F0)/(4 R)$, where T is the surface tension, F is the force required to detach the ring, $F0$ is the buoyancy force of the ring, and R is the radius of the ring.

These are some of the common methods used to determine the surface tension of a given liquid. The choice of method depends on the specific requirements and conditions of the experiment.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly measured using a device called a viscometer. Here are the steps to determine the viscosity of a given liquid using a viscometer:

1. **Select the appropriate viscometer:** Different types of viscometers are available for different types of liquids and measurement ranges. Choose the one that is suitable for the liquid you want to measure.

2. **Calibrate the viscometer:** Before taking any measurements, it is important to calibrate the viscometer to ensure accurate results. Follow the manufacturer's instructions for calibration.

3. **Prepare the sample:** The liquid sample should be free of air bubbles and at a consistent temperature. Pour the liquid into the viscometer carefully to avoid introducing air bubbles.

4. **Take the measurement:** Follow the manufacturer's instructions for taking the measurement. This typically involves timing the flow of the liquid through a narrow tube or capillary.

5. **Calculate the viscosity:** Use the data from the measurement and the calibration to calculate the viscosity of the liquid. The calculation method will vary depending on the type of viscometer used.

6. **Record and report the results:** Record the results of the measurement and report them in the appropriate format. Be sure to include any

relevant information such as the temperature of the liquid and the type of viscometer used.

These are the basic steps for determining the viscosity of a given liquid using a viscometer. It is important to follow the manufacturer's instructions and use the appropriate equipment to ensure accurate results.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is (NH4)2Fe(SO4)2.6H2O.
3. It is commonly used as a standard solution in redox titrations due to its stability and ease of preparation.
4. The strength of a ferrous ammonium sulfate solution can be determined by titrating it against a standard solution of potassium permanganate (KMnO4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide (K3Fe(CN)6).
6. The end point of the titration is indicated by the appearance of a blue color due to the formation of Prussian blue complex.
7. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the balanced chemical equation for the reaction.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is diphenylamine sulfonic acid.
4. The solution of potassium dichromate is titrated against the solution whose strength is to be determined.
5. The end point is indicated by the appearance of a blue color due to the formation of a blue complex between diphenylamine sulfonic acid and chromium ions.
6. The strength of the solution can be calculated using the formula: strength = (normality of potassium dichromate x volume of potassium dichromate used) / volume of solution titrated.
7. This method is accurate and reliable for the determination of the strength of a solution.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula Ca(ClO)2. It is commonly used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

The determination of available chlorine in bleaching powder can be done using the following steps:

1. Weigh a sample of bleaching powder and dissolve it in water.
2. Add excess potassium iodide (KI) to the solution. The iodide ions react with the available chlorine to produce iodine.
3. Titrate the iodine produced with a standard solution of sodium thiosulfate (Na2S2O3) using starch as an indicator. The reaction between iodine and thiosulfate produces iodide ions and sulfate ions.
4. Calculate the amount of available chlorine in the bleaching powder using the volume of thiosulfate solution used in the titration and the concentration of the thiosulfate solution.

This method is known as the iodometric titration method and is commonly used to determine the available chlorine in bleaching powder. It is important to note that the accuracy of the results depends on the purity of the reagents used and the precision of the measurements taken during the experiment.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method known as the Mohr method.
2. The Mohr method involves titrating the water sample with a solution of silver nitrate (AgNO3) using potassium chromate (K2CrO4) as an indicator.
3. The silver nitrate reacts with the chloride ions in the water sample to form a white precipitate of silver chloride (AgCl).
4. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate (Ag2CrO4).
5. The amount of silver nitrate used in the titration can be used to calculate the concentration of chloride ions in the water sample.
6. The Mohr method is a simple and accurate method for determining the chloride content in water samples.
7. It is important to determine the chloride content in water as high levels of chloride can have negative impacts on aquatic life and can also affect the taste of drinking water.
8. The acceptable level of chloride in drinking water is typically less than 250 mg/L.
9. Other methods for determining the chloride content in water include ion chromatography and ion-selective electrodes.

10. It is important to use a reliable and accurate method for determining the chloride content in water to ensure the safety and quality of the water.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol or substituted phenol with formaldehyde. PF resins were the first completely synthetic polymers to be commercialized .

The preparation of PF resin involves a step-growth polymerization reaction that can be either acid- or base-catalyzed. Since formaldehyde exists as a dynamic equilibrium of methylene glycol oligomers for the most part in the solution, the reaction is typically carried out in the presence of a catalyst .

Phenol, formaldehyde, water, and catalyst are mixed in the desired amount, depending on the resin to be formed, and are then heated. The first part of the reaction, at around 70 °C, forms a thick reddish-brown tacky material, which is rich in hydroxymethyl and benzylic ether groups .

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin, also known as urea-methanal, is a non-transparent thermosetting resin or plastic. It is made from urea and formaldehyde when they are heated in the presence of a mild base such as ammonia or pyridine.

The most common method of preparation for commercial UF resin is the addition of a second amount of urea during the reaction. The ratio of urea to formaldehyde is between 1:2 and 1:2.2 and therefore methylolation can take place at in a short amount of time at temperatures between 90 and 95 °C, with a mixture being maintained under reflux.

One way to prepare UF resin under acidic conditions is to mix 5 g of urea with 6 ml of formalin in a test tube, and shake the test tube until the urea has dissolved. Adjust the pH of the solution to 4 by addition of 4 drops of 0.5N H2SO4, and observe the time required for precipitation to occur.

Another way to prepare a urea-formaldehyde adhesive is to charge sixty grams (1 mole) of urea and 137 g (1.7 mole) formalin into a 500-mL reaction kettle equipped with a mechanical stirrer and a reflux condenser. The pH of the mixture is adjusted with 2M NaOH to between 7 and 8 as determined by universal indicator paper.

The synthesis of all UF resins can also be done using a typical three-step procedure (alkaline-acidic-alkaline). First, formaldehyde (F=250 g) is poured into a three-necked flask and adjusted to pH 7.5 to pH 8.0 with 20% sodium hydroxide solution. The first amount of urea (U 1 =94.4 g) is added to give the F/U molar ratio of 2.0.

Urea is manufactured from carbon dioxide and ammonia at a temperature of 135–200 °C and at a pressure of 70–230 atmospheres. Formaldehyde is manufactured by the oxidation of methanol which can be produced from the reaction of carbon dioxide with hydrogen or can be derived from petroleum.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different chemical structures and properties. Here is the information on the preparation of each compound:

Adipic acid:

- Adipic acid is an organic compound with the formula (CH2)4(COOH)2.
- It is a white crystalline powder and is one of the most important dicarboxylic acids.
- Adipic acid is mainly used in the production of nylon.
- It can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
- Another method of preparation is the carbonylation of butadiene in the presence of a nickel catalyst.

Paracetamol:

- Paracetamol, also known as acetaminophen, is an analgesic and antipyretic drug.
- It is commonly used to relieve pain and reduce fever.
- Paracetamol can be prepared by the acetylation of p-aminophenol with acetic anhydride.
- Another method of preparation is the nitration of phenol followed by reduction and acetylation.

14. Determination of Cell Conductance of a solution

Cell conductance is a measure of a solution's ability to conduct an electric current. It is determined by the concentration and mobility of ions in the solution. Here are the steps to determine the cell conductance of a solution:

1. Prepare the solution whose conductance is to be determined.
2. Clean the conductivity cell with distilled water and dry it.
3. Fill the cell with the solution.
4. Connect the cell to a conductivity meter.
5. Record the cell conductance as displayed on the conductivity meter.

It is important to note that the temperature of the solution can affect its conductance. Therefore, it is recommended to measure the temperature of the solution and use a temperature compensation feature if available on the conductivity meter. Additionally, the cell constant of the conductivity cell should be known and entered into the conductivity meter for accurate measurements.

15. Determination of Rate constant of hydrolysis of esters

The rate constant of the hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants. The reaction can be represented by the following equation:

Ester + Water   Acid + Alcohol

The rate of reaction can be determined by measuring the change in concentration of one of the reactants or products over time. This can be done using techniques such as titration or spectrophotometry.

Once the rate of reaction has been determined at different concentrations, the data can be plotted on a graph and the rate constant can be calculated using the appropriate rate law. For a first-order reaction, the rate law is given by:

Rate = k [Ester]

Where k is the rate constant and [Ester] is the concentration of the ester. The rate constant can be determined by plotting the natural logarithm of the concentration of the ester against time and calculating the slope of the line.

For a second-order reaction, the rate law is given by:

Rate = k [Ester] Water

In this case, the rate constant can be determined by plotting 1/[Ester] against time and calculating the slope of the line.

It is important to note that the rate constant is dependent on temperature and the specific ester being used. Therefore, the experiment must be carried out under controlled conditions and the results must be interpreted with care.

16. Element detection and identification of functional groups in organic compounds

- Element detection and identification of functional groups in organic compounds is essential for understanding the structure, characteristics, and interactions of organic molecules that include carbon.
- Functional groups are sites of chemical reactivity in a molecule.
- Detection of functional groups starts from the physical character like color, state, odor, stability, etc. Organic compounds may be crystalline or amorphous.
- Molecules of organic compounds, except for hydrocarbons, can be divided into two parts: a reactive part referred to as a functional group and a skeleton of carbon atoms called an alkyl group. The properties of a compound are largely determined by the functional group.

- Some common functional groups include alcohols, phenols, aldehydes, ketones, carboxylic acids, and amines.
- There are various tests to detect the presence of these functional groups, such as the 2,4-dinitrophenyl hydrazine test for carbonyls (aldehydes and ketones).
- Functional groups have different priorities when it comes to naming. They are assigned priorities based broadly on their reactivity.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions provided by the instructor.
- Students should also be prepared for any changes to the list of experiments.

Course Outcomes:

1. Understanding of the fundamental concepts and principles of the subject.
2. Ability to apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
3. Development of critical thinking and analytical skills.
4. Improvement in communication and collaboration skills.
5. Enhancement of lifelong learning skills and the ability to adapt to new situations and challenges.
6. Acquisition of professional and ethical values and standards.
7. Preparation for further study or employment in the field.

Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the knowledge and skills acquired in the course to solve problems.
3. Communicate effectively in both written and oral forms.
4. Work effectively in a team.
5. Demonstrate critical thinking and analytical skills.
6. Demonstrate ethical and professional behavior.
7. Demonstrate the ability to learn independently and continuously.
8. Demonstrate the ability to adapt to new situations and technologies.
9. Demonstrate the ability to integrate knowledge from different disciplines.
10. Demonstrate the ability to apply knowledge in practical situations.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should possess upon completion of a course. These outcomes are typically aligned with the course objectives and are used to assess student learning.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. The taxonomy consists of six levels, with the lower levels representing basic knowledge and comprehension, and the higher levels representing more advanced cognitive skills such as analysis, synthesis, and evaluation.

The six levels of Bloom's Taxonomy are: 1. **Remembering**: recalling previously learned information. 2. **Understanding**: comprehending the meaning of the information. 3. **Applying**: using the information in a new situation. 4. **Analyzing**: breaking down the information into its component parts to understand its structure. 5. **Evaluating**: making judgments about the value of the information. 6. **Creating**: combining the information to form a new whole.

By aligning course outcomes with Bloom's Taxonomy, instructors can ensure that their course is designed to promote higher-order thinking skills and that assessments are appropriately challenging for students. This can help students to develop a deeper understanding of the course material and to apply their knowledge in new and meaningful ways.

Level

- A level is a tool used to determine if a surface is horizontal or vertical.
- It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
- When the level is placed on a surface, the bubble will move to the highest point of the tube, indicating the slope of the surface.
- If the bubble is centered between two marked lines on the tube, the surface is level.
- Levels come in various sizes and shapes, including torpedo, I-beam, and box beam levels.
- They are commonly used in construction, carpentry, and surveying, among other fields.
- Laser levels and digital levels are more advanced types of levels that can provide more precise measurements.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are devices used to measure, analyze, and quantify the physical and chemical properties of a sample. These instruments are commonly used in scientific research, quality control, and medical diagnosis. Some of the most commonly used analytical instruments include:

1. **Spectrophotometers**: These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs**: These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers**: These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical instruments**: These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Microscopes**: These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations, and the choice of instrument depends on the specific needs of the analysis. Understanding the use of different analytical instruments is essential for selecting the appropriate instrument for a given task and for interpreting the results obtained from the instrument.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it is covered by an elastic membrane. Surface tension is typically measured in units of force per unit length, such as millinewtons per meter (mN/m).

There are several methods for measuring surface tension, including:

1. **Maximum bubble pressure method**: This method involves injecting gas into a liquid through a small tube and measuring the pressure required to form a bubble at the end of the tube. The surface tension is calculated from the maximum pressure required to form the bubble.
2. **Drop weight method**: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated from the weight of the droplet and the radius of the tube.

3. **Wilhelmy plate method**: This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it from the liquid. The surface tension is calculated from the force and the dimensions of the plate.

4. **Du Nouy ring method**: This method involves immersing a ring into a liquid and measuring the force required to detach it from the liquid. The surface tension is calculated from the force and the dimensions of the ring.

These are some of the common methods used to measure surface tension. Each method has its own advantages and limitations, and the choice of method may depend on factors such as the type of liquid being measured and the accuracy required.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

1. **Viscosity** is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the cohesive forces between the molecules. The viscosity of a fluid can be affected by factors such as temperature, pressure, and the presence of dissolved solids.

2. **Conductance of a solution** is a measure of its ability to conduct electricity. The conductance of a solution is determined by the concentration and mobility of the ions present in the solution. The more ions present in the solution, the higher its conductance.

3. **Chloride ions** are approximately 1.9 percent of the mass of seawater. The more chloride ions present in the water, the higher the conductivity. In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm .

4. **Iron content** in water can affect its taste, color, and odor. High levels of iron in water can also cause staining of clothes and plumbing fixtures. Iron can be present in water in two forms: ferrous iron (Fe^{2+}) and ferric iron (Fe^{3+}). Ferrous iron is soluble in water, while ferric iron is insoluble and can cause discoloration.

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3, a Belgian-Dutch girl group formed in 1998.
2. K3, a mountain in the Karakoram range, also known as Broad Peak.
3. K3, a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.

4. K3, a type of surface in algebraic geometry.
5. K3, a type of potassium channel.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters to measure in water. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the water's ability to neutralize acids.

1. **Measuring Hardness**: Hardness can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to cause the color change is used to calculate the hardness of the water. Alternatively, hardness test kits are available that use colorimetric methods to determine hardness.

2. **Measuring Alkalinity**: Alkalinity can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to cause the color change is used to calculate the alkalinity of the water. Alternatively, alkalinity test kits are available that use colorimetric methods to determine alkalinity.

It is important to measure both hardness and alkalinity in water to ensure that it is suitable for its intended use. For example, water with high hardness can cause scaling in pipes and appliances, while water with low alkalinity can be corrosive.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion ($H+$) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. From Haloarenes: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .

2. From Benzene Sulphonic Acid: Sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide at 573 – 623K. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.

3. From Cumene: Phenol can be prepared in large quantities by the air oxidation of cumene.

4. From Diazonium Salts: Phenol can be prepared in small amounts by the hydrolysis of diazonium salts.

5. From Coal Tar: Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.

6. Raschig's Process: The reaction of benzene, hydrogen chloride, and air at 523K produces phenol.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- UF is synthesized using the "alkali-acid-alkali" method and often shows the appearance of crystal domains .
- UF resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Adipic acid dihydrazide can be used as a modifier to reduce the free formaldehyde in UF resin.

K3

- K3 is a term that can refer to several different things, depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a type of visa for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- Without more context, it is difficult to determine which specific meaning of K3 is being referred to.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted as k, is a measure of the speed at which a chemical reaction occurs. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law of the reaction.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by conducting experiments and analyzing the data.
2. Write the rate law for the reaction using the determined orders of the reactants.
3. Conduct experiments to measure the initial rate of the reaction for different initial concentrations of the reactants.

4. Substitute the experimental data into the rate law to solve for the rate constant, k.

It is important to note that the rate constant is temperature-dependent and its value will change with changes in temperature. The Arrhenius equation can be used to determine the temperature dependence of the rate constant.

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1. I'm sorry, but I don't have enough information to write about the topic "Page 33 of 40". Could you please provide more context or details about the topic you would like me to write about?
2. Is there a specific book or document that you are referring to?
3. What is the subject matter or theme of the content on page 33 of 40?

ENGINEERING GRAPHICS LAB KCS

Engineering Graphics Lab KCS is a course that teaches students the fundamentals of technical drawing and computer-aided design (CAD). The course covers topics such as:

1. Orthographic projection: This is a method of representing three-dimensional objects in two dimensions. It is commonly used in engineering and architecture to create detailed technical drawings.

2. Isometric projection: This is another method of representing three-dimensional objects in two dimensions. It is commonly used in technical illustrations and diagrams.

3. Dimensioning and tolerancing: This is the process of specifying the size and geometric characteristics of an object. It is an essential part of technical drawing and is used to ensure that parts fit together correctly.

4. CAD software: This is a type of software used to create, modify, analyze, and optimize designs. It is commonly used in engineering, architecture, and product design.

5. 3D modeling: This is the process of creating a digital representation of a physical object. It is commonly used in product design, animation, and video game development.

The course is designed to provide students with the skills and knowledge needed to create technical drawings and designs using both traditional and computer-based methods. It is a valuable course for students pursuing careers in engineering, architecture, and product design.

Course Objectives

- To provide a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To encourage independent learning and research.
- To foster effective communication and collaboration skills.
- To prepare students for further academic or professional pursuits in the field.
- To promote the application of theoretical knowledge to real-world scenarios.
- To instill a sense of ethical responsibility and social awareness.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in scientific research, medical diagnosis, and quality control in various industries.

Here are some fundamental concepts of analytical instruments:

1. **Accuracy**: This refers to how close the measurement is to the true value of the substance being analyzed. Analytical instruments must be accurate to provide reliable results.

2. **Precision**: This refers to the consistency of the measurements. An instrument that provides precise results will give the same measurement when the same substance is analyzed multiple times.

3. **Sensitivity**: This refers to the ability of the instrument to detect small changes in the substance being analyzed. A sensitive instrument can detect even minute changes in the concentration of a substance.

4. **Selectivity**: This refers to the ability of the instrument to distinguish between different substances. A selective instrument can accurately identify and measure the concentration of a specific substance in a mixture of substances.

5. **Calibration**: This is the process of adjusting the instrument to ensure that it provides accurate and precise measurements. Calibration is essential to ensure the reliability of the results obtained from the instrument.

6. **Maintenance**: Regular maintenance is necessary to ensure the proper functioning of the instrument. This includes cleaning, calibration, and replacement of worn-out parts.

These are some of the fundamental concepts of analytical instruments that students should understand. A thorough understanding of these concepts will enable students to use analytical instruments effectively and obtain reliable results.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content**: Chloride content refers to the amount of chloride ions present in a water sample. It is an important parameter to measure as high levels of chloride can indicate contamination from sources such as sewage, industrial waste, or road salt. The chloride content of water can be measured using various methods, including titration, ion chromatography, and ion-selective electrodes.
2. **Hardness**: Hardness refers to the concentration of calcium and magnesium ions in water. These ions can cause problems such as scaling in pipes and reduced effectiveness of soap and detergents. Hardness can be measured using titration, atomic absorption spectroscopy, or ion exchange chromatography.
3. **Alkalinity**: Alkalinity refers to the ability of water to neutralize acids. It is an important parameter to measure as it can affect the pH of water and the effectiveness of water treatment processes. Alkalinity can be measured using titration, colorimetry, or ion chromatography.

In summary, the analysis of chloride content, hardness, and alkalinity are important parameters to measure in water quality testing. Various methods can be used to measure these parameters, and the results can provide valuable information about the quality of the water sample.

Water

Water is a transparent, tasteless, odorless, and nearly colorless chemical substance, which is the main constituent of Earth's streams, lakes, and oceans, and the fluids of most living organisms. It is vital for all known forms of life, even though it provides no calories or organic nutrients.

- Water is a liquid at standard temperature and pressure.
- Water covers 71% of the Earth's surface, mostly in seas and oceans.
- Small portions of water occur as groundwater (1.7%), in the glaciers and the ice caps of Antarctica and Greenland (1.7%), and in the air as vapor, clouds (formed of ice and liquid water suspended in air), and precipitation (0.001%).
- Water plays an important role in the world economy.
- Approximately 70% of the freshwater used by humans goes to agriculture.
- Water is a good solvent for a wide variety of chemical substances; as such it is widely used in industrial processes, and in cooking and washing.

- Water is essential to the survival of all known forms of life.
- Water is a transparent and nearly colorless chemical substance that is the main constituent of Earth's streams, lakes, and oceans, and the fluids of most living organisms.
- Water is a polar molecule, meaning that it has a partial positive charge near the hydrogen atoms and a partial negative charge near the oxygen atom.
- Water is a good solvent for many ionic compounds, such as salts, acids, and bases.
- Water is also a good solvent for many polar molecules, such as sugars and alcohols.
- Water is a poor solvent for nonpolar molecules, such as oils and fats.
- Water has a high heat capacity, meaning that it can absorb a large amount of heat before its temperature rises.
- Water has a high heat of vaporization, meaning that it takes a large amount of heat to turn water into steam.
- Water has a high surface tension, meaning that it tends to stick to itself and form droplets.
- Water is a good conductor of heat, meaning that it can transfer heat from one place to another.
- Water is a poor conductor of electricity, meaning that it does not easily allow electric current to flow through it.
- Water is a reactive chemical substance, meaning that it can react with many other substances to form new compounds.
- Water is a renewable resource, meaning that it can be replenished naturally over time.
- Water is a finite resource, meaning that there is a limited amount of water available on Earth.
- Water is a precious resource, meaning that it is valuable and should be used wisely.
- Water is a shared resource, meaning that it belongs to everyone and should be managed for the benefit of all.

Measure of pH, Surface Tension and Viscosity

pH

- pH is a measure of the acidity or basicity of a solution.
- It is defined as the negative logarithm of the hydrogen ion concentration in a solution.
- The pH scale ranges from 0 to 14, with 7 being neutral.
- A pH value less than 7 indicates an acidic solution, while a pH value greater than 7 indicates a basic solution.

Surface Tension

- Surface tension is the property of a liquid that allows it to resist an external force.
- It is caused by the cohesive forces between the molecules of the liquid.
- Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
- The surface tension of a liquid can be affected by various factors, including temperature, impurities, and the presence of surfactants.

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- It is defined as the ratio of the shear stress to the shear rate in a fluid.
- Viscosity is typically measured in units of pascal-seconds ($Pa \cdot s$).
- The viscosity of a fluid can be affected by various factors, including temperature, pressure, and the presence of impurities.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to conform to the shape of its container, and its ability to flow.
- The particles in a liquid are close together and are constantly moving, allowing them to slide past one another.
- The attractive forces between the particles in a liquid are strong enough to keep the particles close together, but not strong enough to keep them in a fixed position.
- Liquids have a definite volume, but not a definite shape.
- The surface of a liquid is always level, due to the force of gravity.
- Liquids can be poured and can take the shape of their container.
- The viscosity of a liquid is a measure of its resistance to flow.
- The boiling point of a liquid is the temperature at which its vapor pressure is equal to the atmospheric pressure.
- The freezing point of a liquid is the temperature at which it turns into a solid.
- Liquids can be mixed with other liquids to form solutions.
- Some liquids, such as water, are essential for life.

Preparation of Different Resins

Resins can be derived from various sources such as polymers, monomers, oils, fatty acids, and natural sources. They have a wide range of applications in industries such as paints and coatings, personal care and home care, pharmaceuticals, rubber, plastics, electrical and jewelry articles, decorative items, wood coatings, etc.

There are two categories of resinous products: natural resins and prepared resins. So far, no general method has been suggested or proposed for the preparation of resins .

In resin casting, fluid synthetic resin is blended with a curing agent, commonly at room temperature or close to room temperature. The two substances are then poured into a mold cavity. The curing agent then converts the resin into solid polymers, essentially solidifying it .

There are several types of resins, including epoxy resin, UV resin, polyester resin, and polyurethane resin. The majority of resins are made up of two elements: the base resin and a catalyst (which is a hardener). Combining the two elements results in a chemical reaction that allows the resin to set .

By manipulating the structure of the linker, resins ranging from extremely acid-labile to base-labile can be prepared. Using linkers additionally allows the preparation of resins with special applications such as DHP resin utilized as a solid-phase support for alcohols or Weinreb resin utilized for preparing aldehydes and ketones .

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid is a dicarboxylic acid with the chemical formula C6H10O4.
2. It can be synthesized from a variety of starting materials, including cyclohexanol, cyclohexanone, and benzene.
3. One common method for synthesizing adipic acid is through the oxidation of cyclohexanol or cyclohexanone using nitric acid.
4. Another method involves the oxidation of benzene using a mixture of nitric and sulfuric acids.
5. The resulting adipic acid can be purified through recrystallization or other separation techniques.

These are some of the key points to understand about the synthesis of adipic acid. It is an important industrial chemical with a variety of uses and can be synthesized through several different methods.

Acid and Paracetamol

- Mefenamic Acid + Paracetamol is a combination of two medicines: Mefenamic Acid and Paracetamol. It works by blocking the release of certain chemical messengers that cause fever, pain and inflammation (redness and swelling).

- Mefenamic Acid+Paracetamol is used for pain relief and fever. It relieves pain and inflammation in conditions like rheumatoid arthritis, osteoarthritis and menstrual pain. It is also used to treat mild to moderate pain.
- Paracetamol may be made by acetylation of para-aminophenol (obtained by reduction of para-nitrophenol) with acetic acid or acetic anhydride.
- Paracetamol is used as an analgesic and antipyretic drug. It is the preferred alternative analgesic-antipyretic to aspirin (acetylsalicylic acid), particularly in patients with coagulation disorders, individuals with a history of peptic ulcer or who cannot tolerate aspirin, as well as in children.
- Paracetamol (Panadol, Calpol, Alvedon) is an analgesic and antipyretic drug that is used to temporarily relieve mild-to-moderate pain and fever. It is commonly included as an ingredient in cold and flu medications and is also used on its own. Paracetamol is exactly the same drug as acetaminophen (Tylenol).